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(54) **IMAGE CAPTURING APPARATUS, CONTROL METHOD, AND STORAGE MEDIUM**

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(57) **ABSTRACT**

A setting unit sets specific information. An association unit associates the set specific information with an image that has been shot using an image capturing unit. In a case where a medical mode for shooting a medical image is ON, a control unit performs control to in a case where the specific information has not been set, cause an image capturing apparatus to transition to a power saving state when a state where a user has not operated the image capturing apparatus has continued for a predetermined time period, and in a case where the specific information has been set, not cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

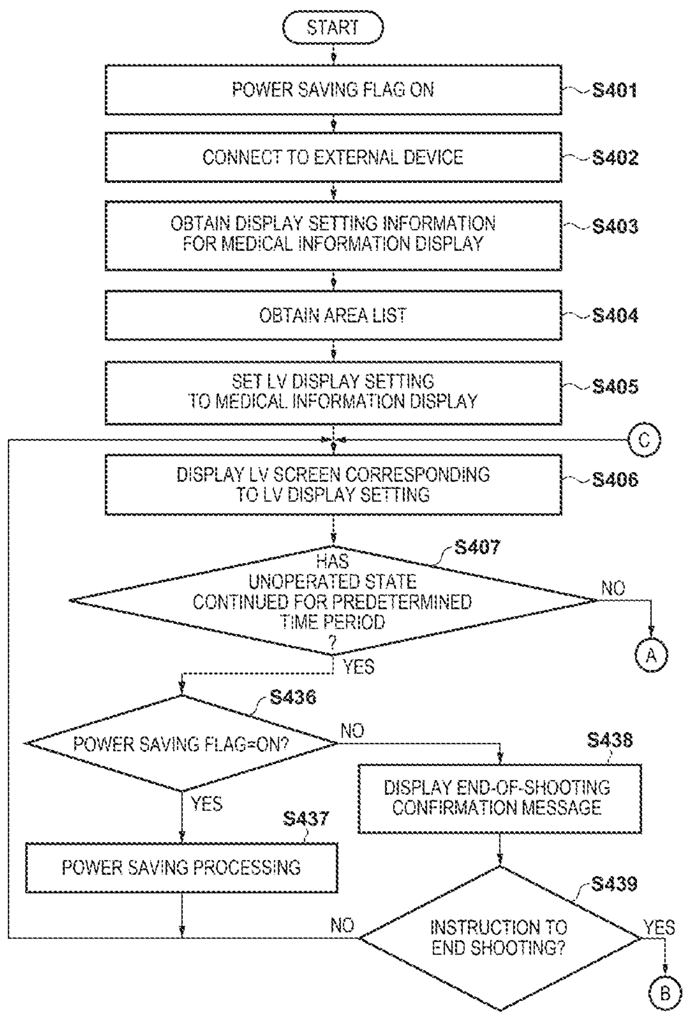


FIG. 1A

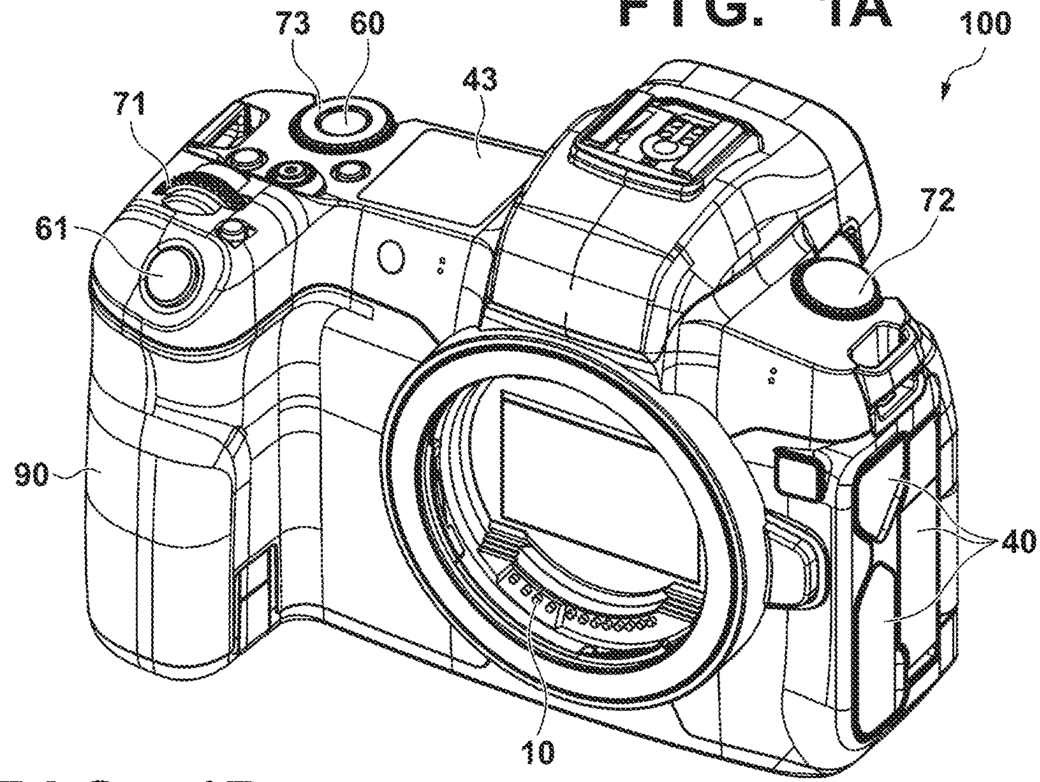
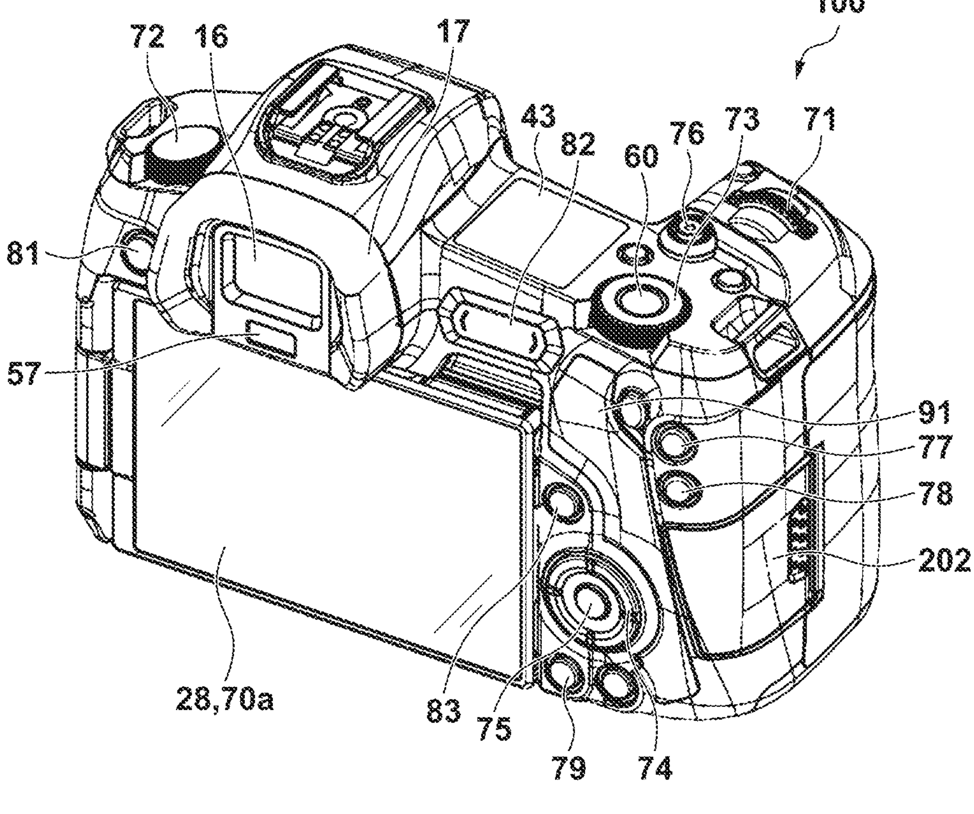
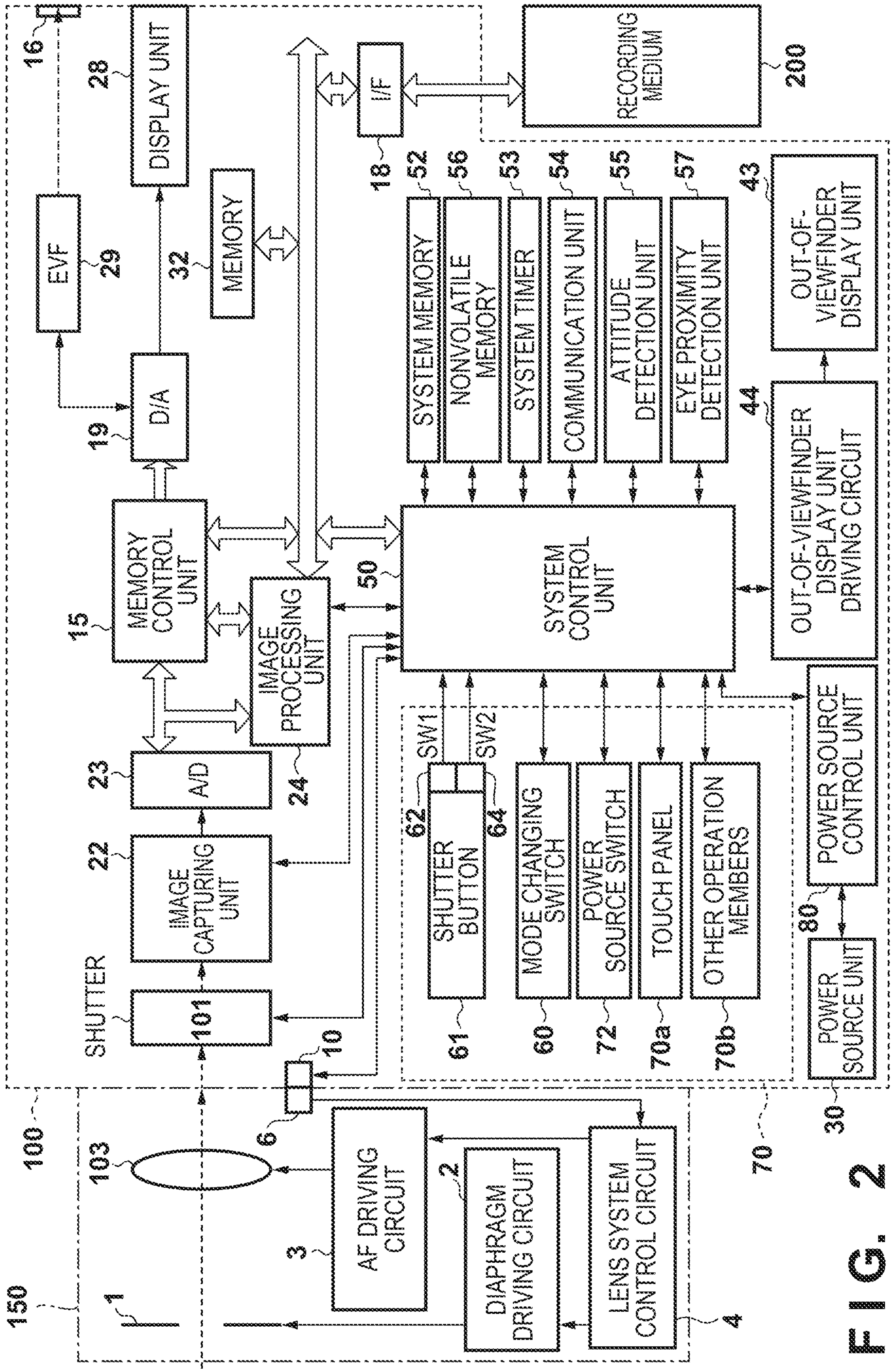

100
73
60
43
71
72
61
90
40
10

FIG. 1B


100
72
16
17
43
82
60
76
73
71
81
57
91
77
78
202
28,70a
83
75
79
74



150
1
100
103
SHUTTER
101
IMAGE CAPTURING UNIT
22
A/D
23
MEMORY CONTROL UNIT
15
D/A
19
EVF
29
16
DISPLAY UNIT
28
MEMORY
32
IMAGE PROCESSING UNIT
24
I/F
18
RECORDING MEDIUM
200
SYSTEM CONTROL UNIT
50
SYSTEM MEMORY
52
NONVOLATILE MEMORY
56
SYSTEM TIMER
53
COMMUNICATION UNIT
54
ATTITUDE DETECTION UNIT
55
EYE PROXIMITY DETECTION UNIT
57
OUT-OF-VIEWFINDER DISPLAY UNIT
43
OUT-OF-VIEWFINDER DISPLAY UNIT DRIVING CIRCUIT
44
SW1
SW2
62
64
SHUTTER BUTTON
61
MODE CHANGING SWITCH
60
POWER SOURCE SWITCH
72
TOUCH PANEL
70a
OTHER OPERATION MEMBERS
70b
70
POWER SOURCE CONTROL UNIT
80
POWER SOURCE UNIT
30
10
6
AF DRIVING CIRCUIT
3
DIAPHRAGM DRIVING CIRCUIT
2
LENS SYSTEM CONTROL CIRCUIT
4

FIG. 2

FIG. 3A

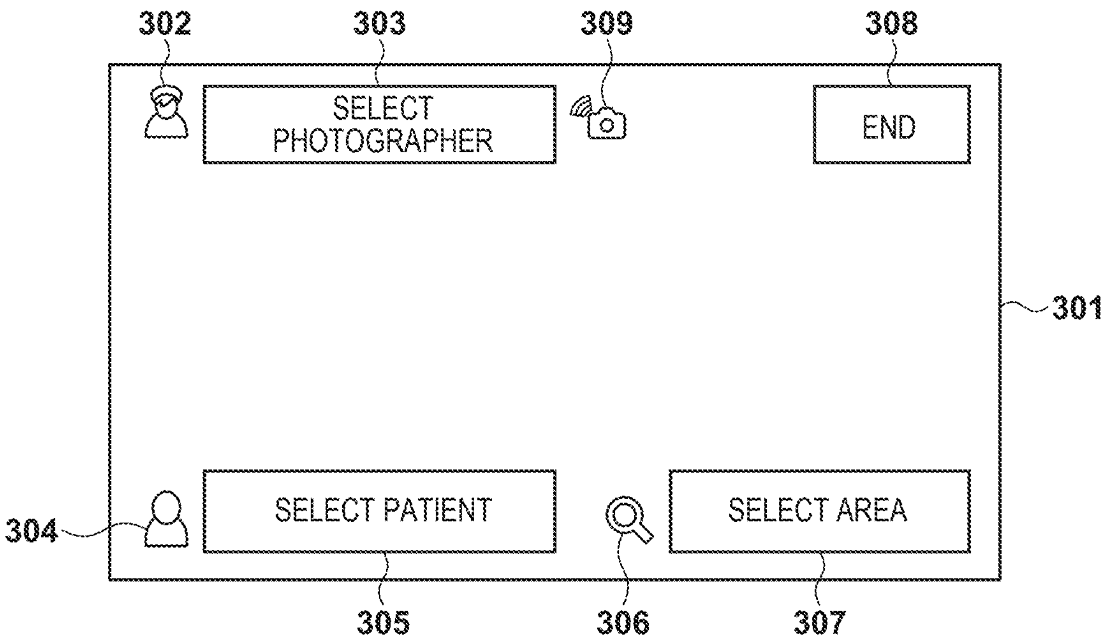

302
303
309
308
SELECT PHOTOGRAPHER
END
301
304
SELECT PATIENT
SELECT AREA
305
306
307

FIG. 3B

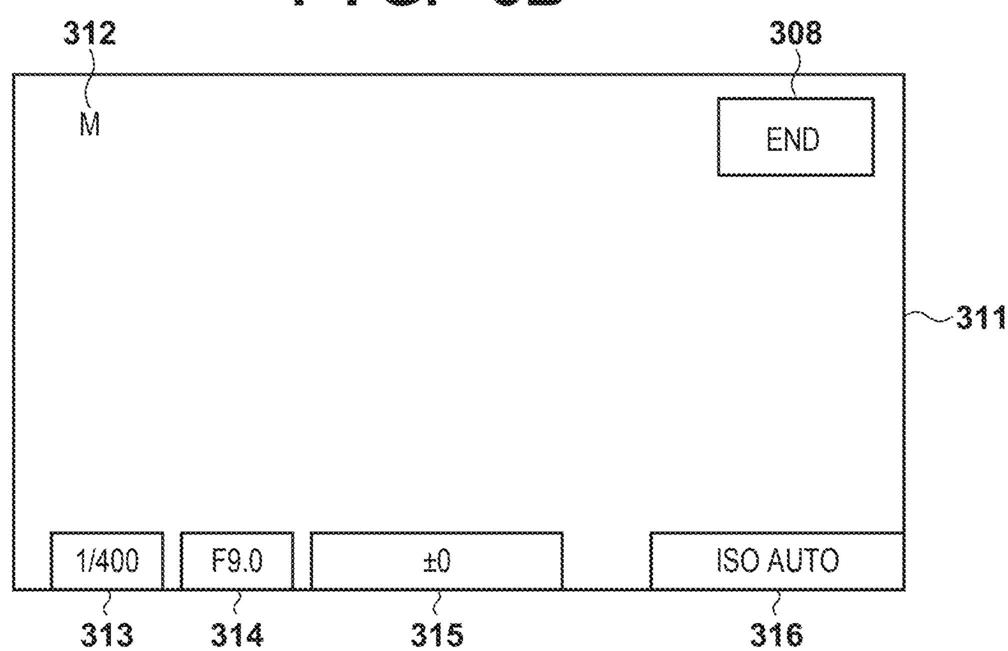

312
308
M
END
311
1/400
F9.0
±0
ISO AUTO
313
314
315
316

FIG. 4A

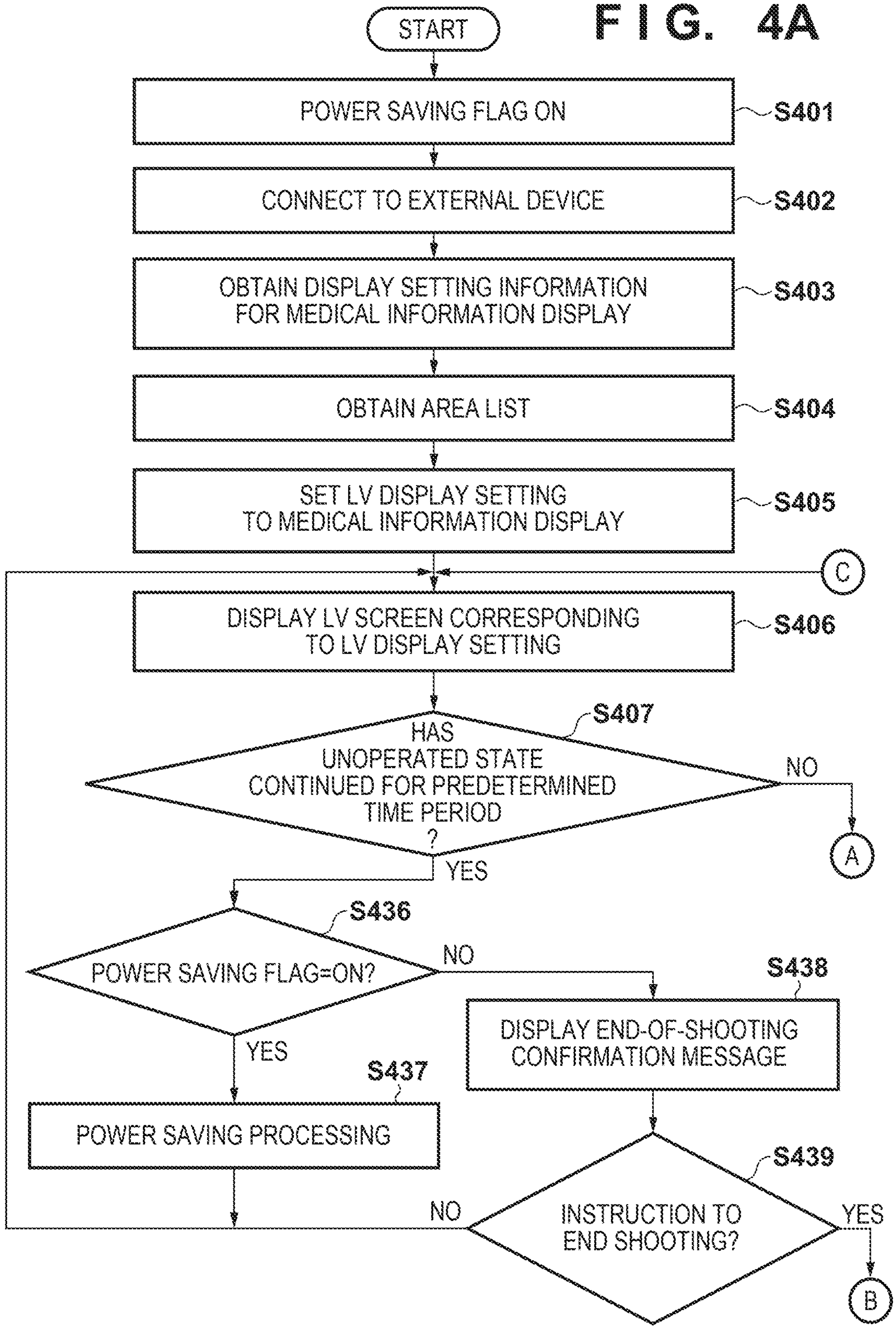
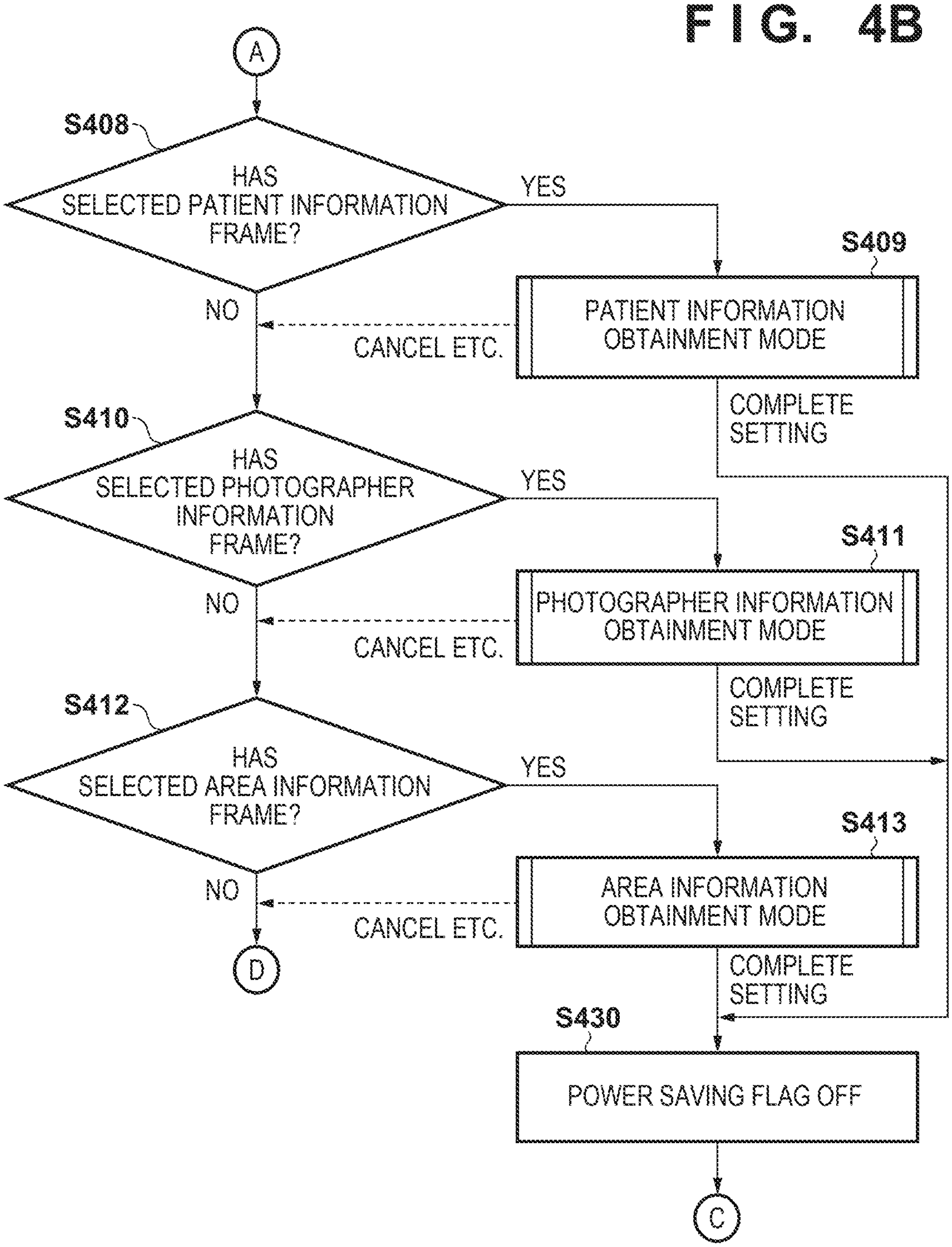

START
POWER SAVING FLAG ON
S401
CONNECT TO EXTERNAL DEVICE
S402
OBTAIN DISPLAY SETTING INFORMATION FOR MEDICAL INFORMATION DISPLAY
S403
OBTAIN AREA LIST
S404
SET LV DISPLAY SETTING TO MEDICAL INFORMATION DISPLAY
S405
C
DISPLAY LV SCREEN CORRESPONDING TO LV DISPLAY SETTING
S406
HAS UNOPERATED STATE CONTINUED FOR PREDETERMINED TIME PERIOD ?
S407
NO
A
YES
POWER SAVING FLAG=ON?
S436
NO
YES
DISPLAY END-OF-SHOOTING CONFIRMATION MESSAGE
S438
POWER SAVING PROCESSING
S437
INSTRUCTION TO END SHOOTING?
S439
NO
YES
B

FIG. 4B


A
S408
HAS SELECTED PATIENT INFORMATION FRAME?
YES
NO
S409
PATIENT INFORMATION OBTAINMENT MODE
CANCEL ETC.
COMPLETE SETTING
S410
HAS SELECTED PHOTOGRAPHER INFORMATION FRAME?
YES
NO
S411
PHOTOGRAPHER INFORMATION OBTAINMENT MODE
CANCEL ETC.
COMPLETE SETTING
S412
HAS SELECTED AREA INFORMATION FRAME?
YES
NO
S413
AREA INFORMATION OBTAINMENT MODE
CANCEL ETC.
COMPLETE SETTING
D
S430
POWER SAVING FLAG OFF
C

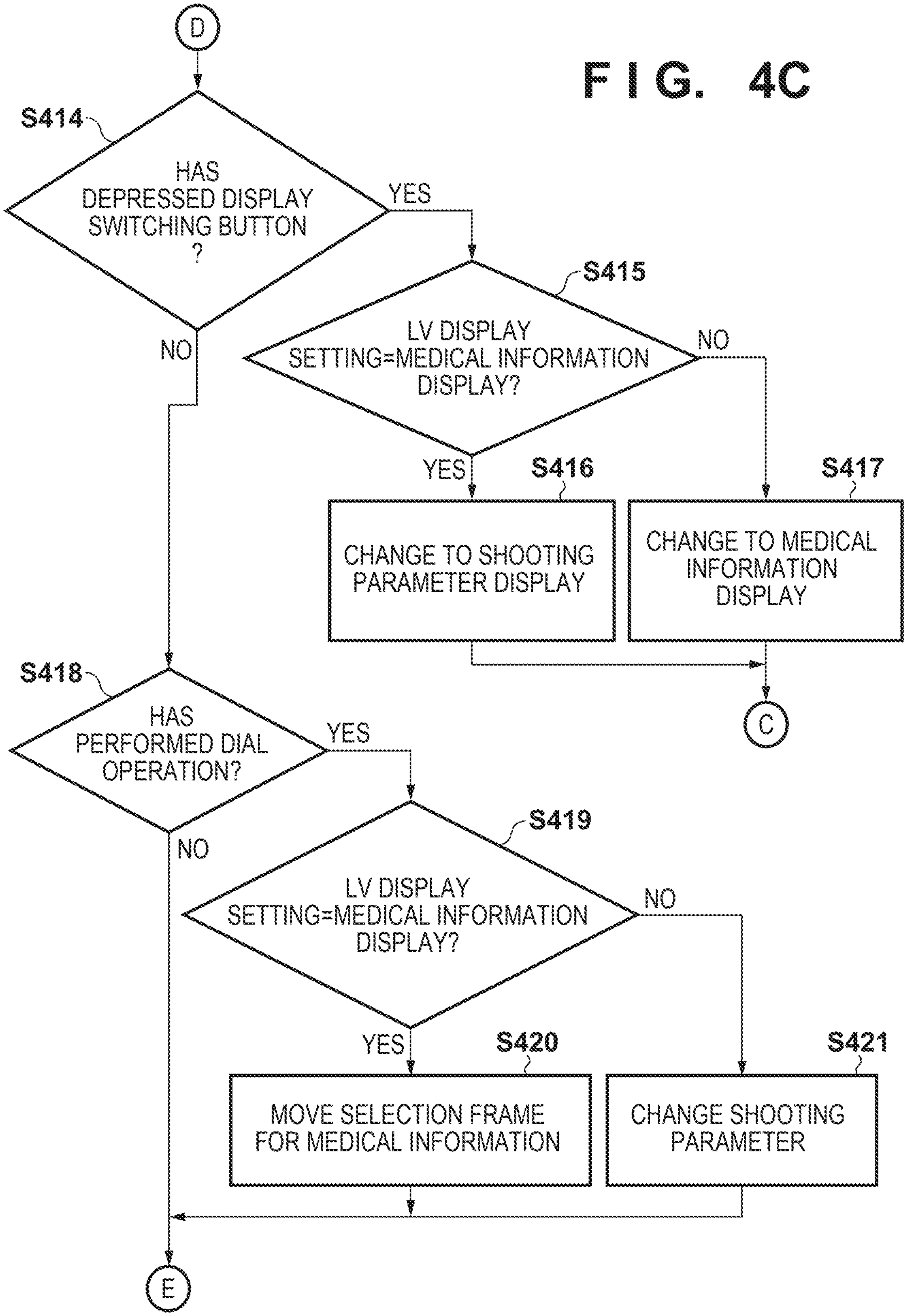

D
S414
HAS DEPRESSED DISPLAY SWITCHING BUTTON ?
YES
NO
S415
LV DISPLAY SETTING=MEDICAL INFORMATION DISPLAY?
NO
YES
S416
CHANGE TO SHOOTING PARAMETER DISPLAY
S417
CHANGE TO MEDICAL INFORMATION DISPLAY
C
S418
HAS PERFORMED DIAL OPERATION?
YES
NO
S419
LV DISPLAY SETTING=MEDICAL INFORMATION DISPLAY?
NO
YES
S420
MOVE SELECTION FRAME FOR MEDICAL INFORMATION
S421
CHANGE SHOOTING PARAMETER
E

FIG. 4C

FIG. 4D

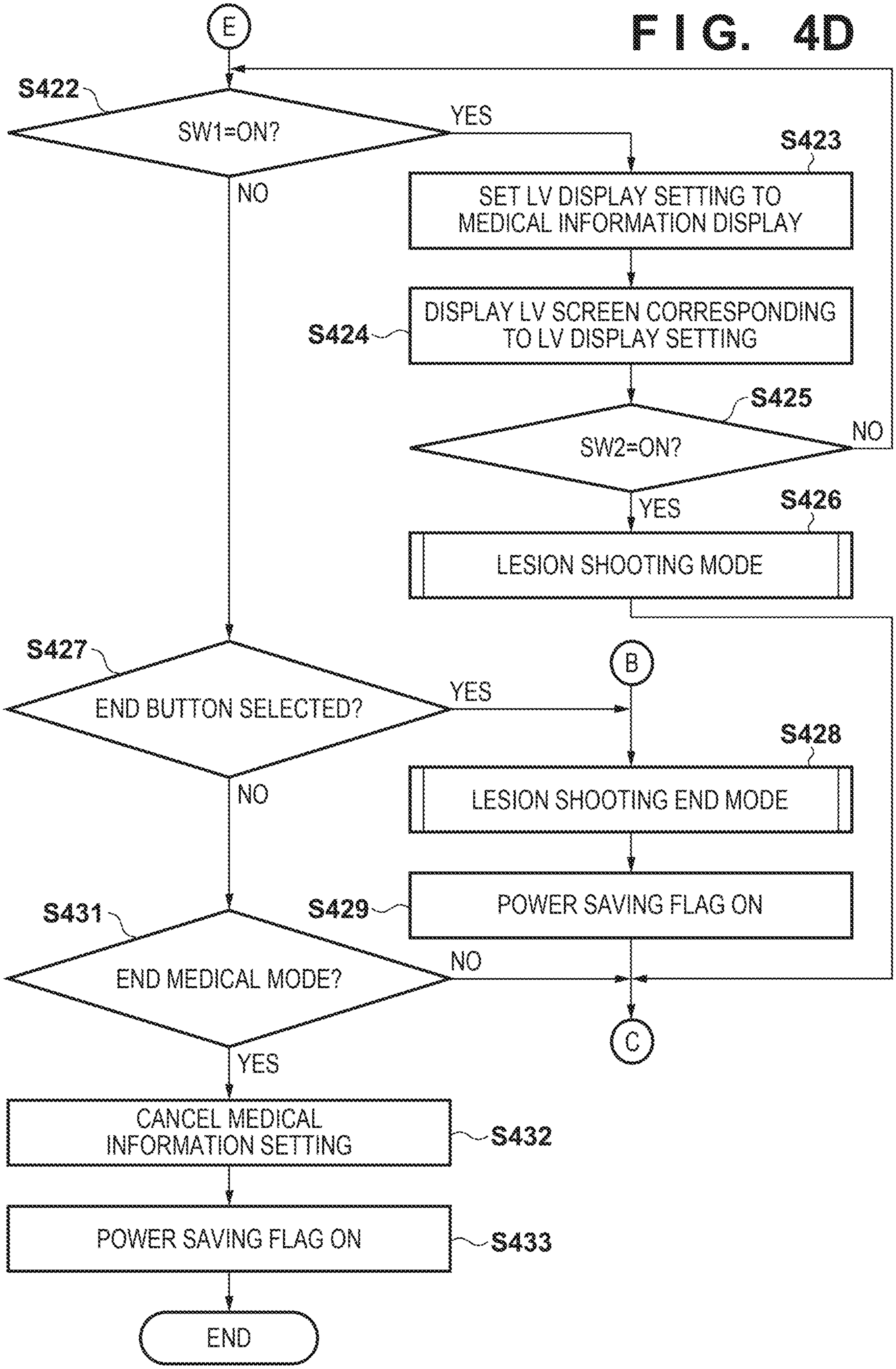

E
S422
SW1=ON?
YES
NO
S423
SET LV DISPLAY SETTING TO MEDICAL INFORMATION DISPLAY
S424
DISPLAY LV SCREEN CORRESPONDING TO LV DISPLAY SETTING
S425
SW2=ON?
NO
YES
S426
LESION SHOOTING MODE
S427
END BUTTON SELECTED?
YES
B
NO
S428
LESION SHOOTING END MODE
S429
POWER SAVING FLAG ON
S431
END MEDICAL MODE?
NO
C
YES
CANCEL MEDICAL INFORMATION SETTING
S432
POWER SAVING FLAG ON
S433
END

FIG. 5A

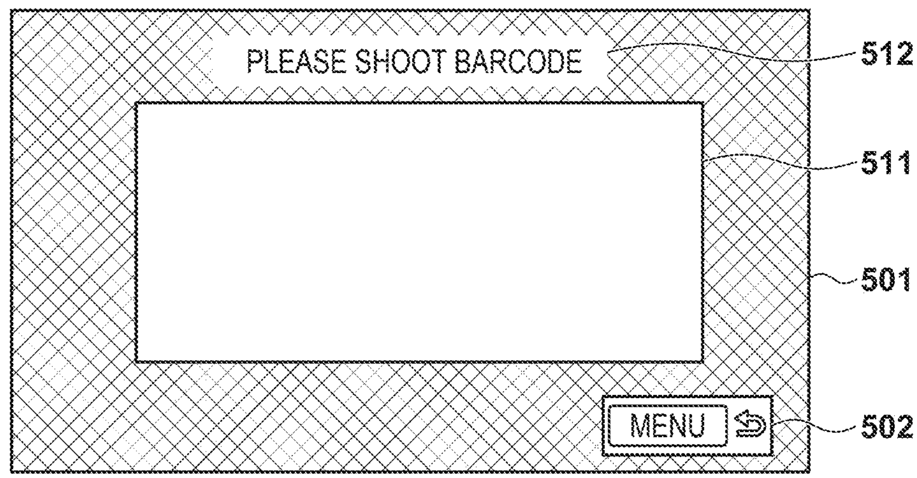


FIG. 5B

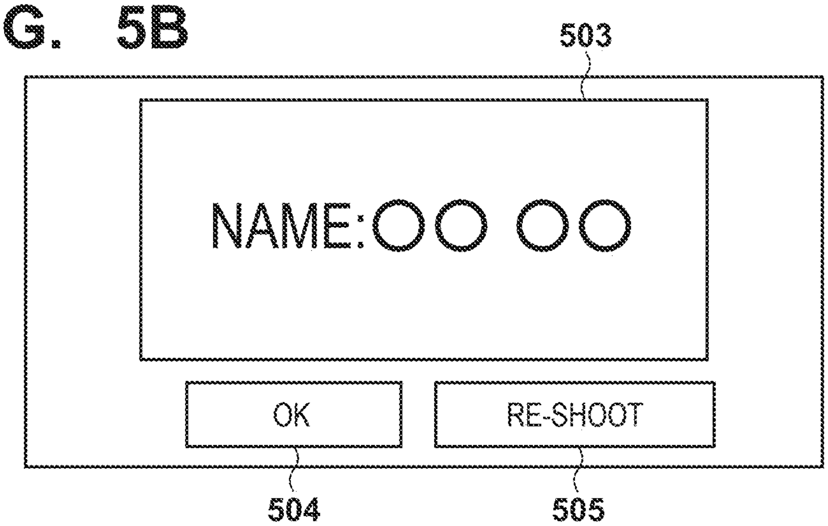


FIG. 5C

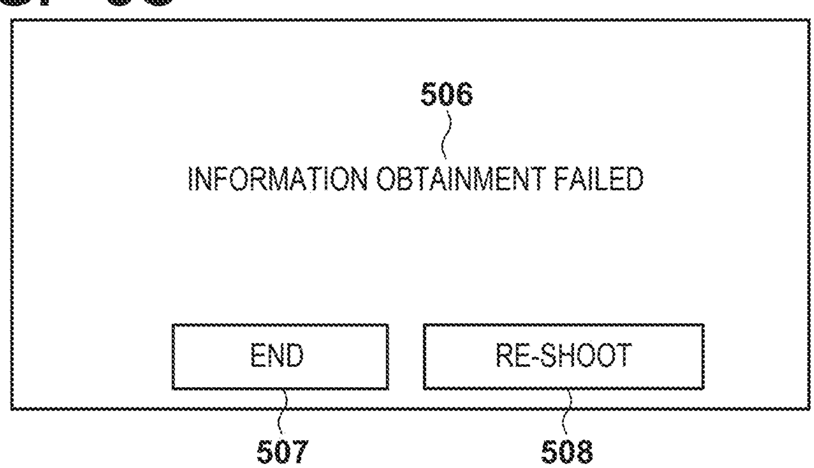


FIG. 6A

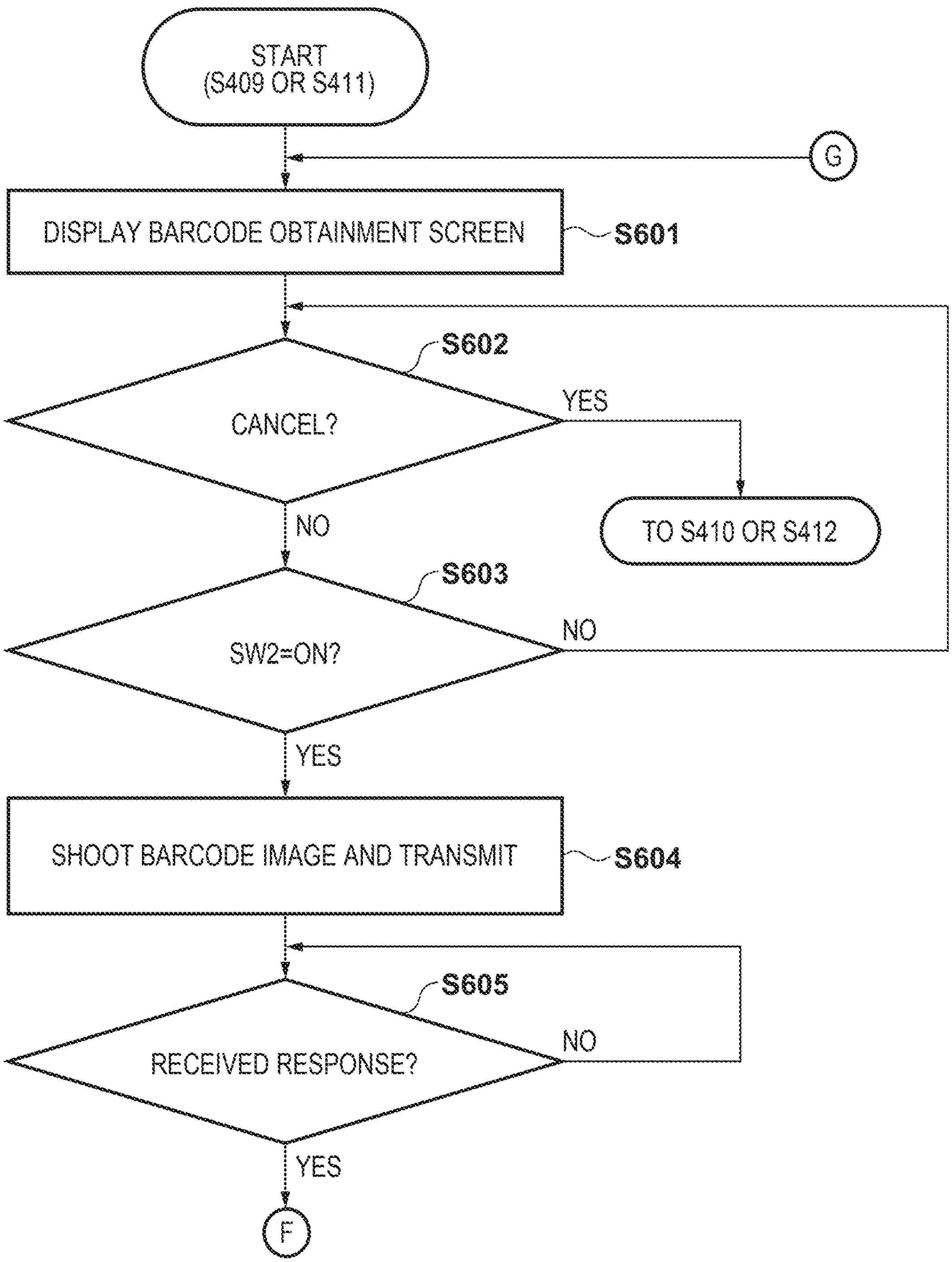
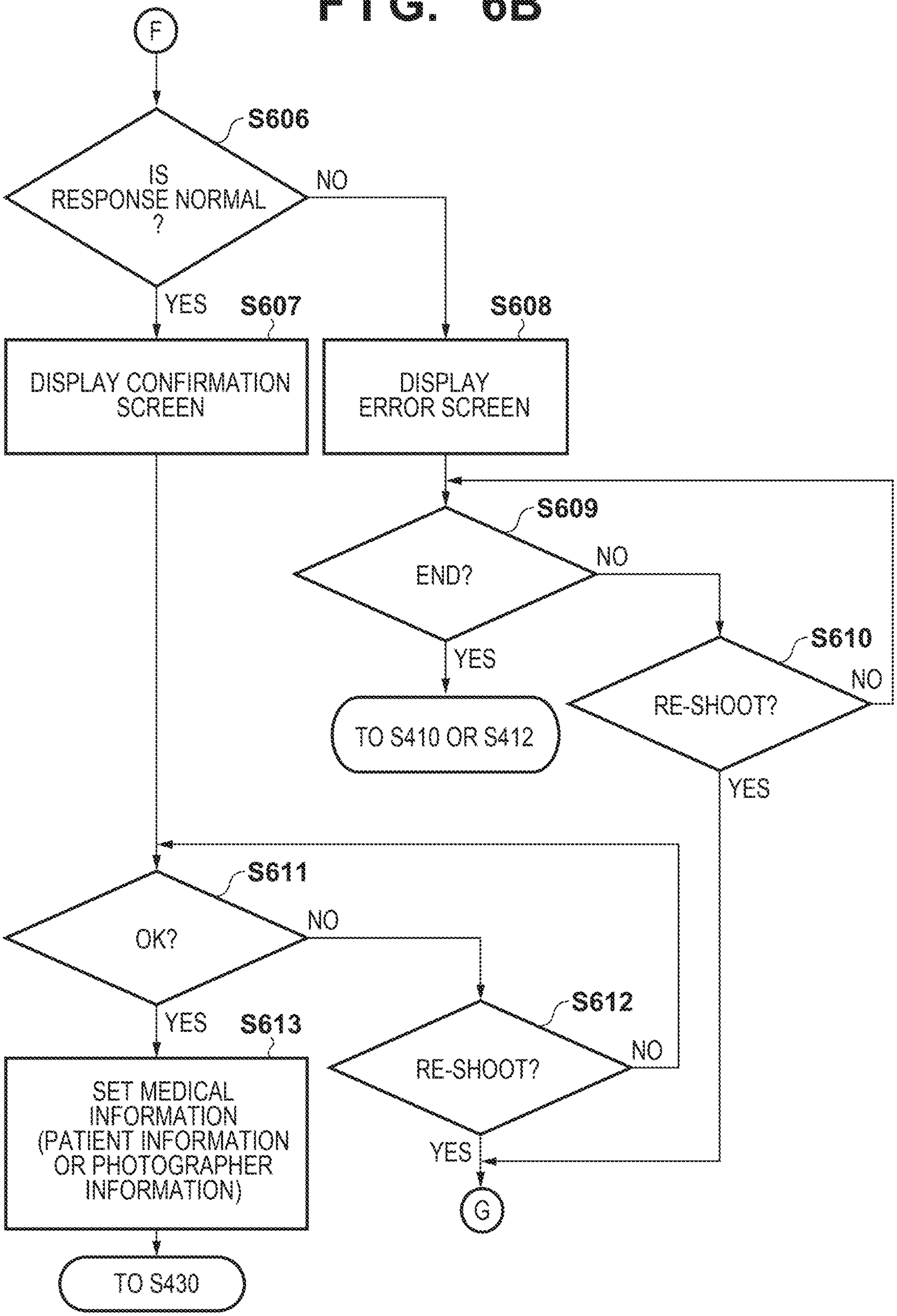
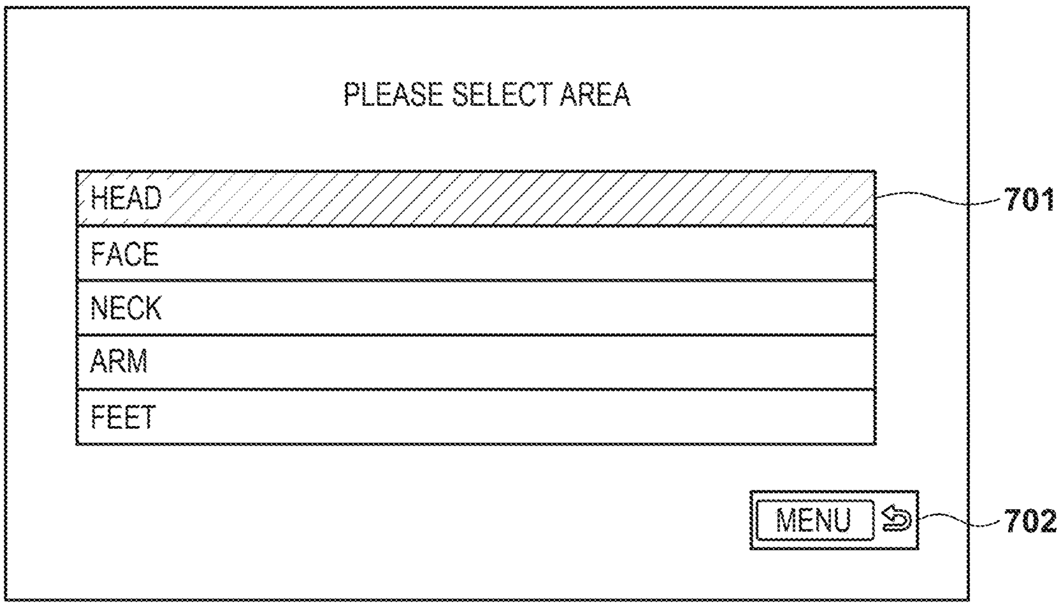

START
(S409 OR S411)
G
DISPLAY BARCODE OBTAINMENT SCREEN
S601
S602
CANCEL?
YES
TO S410 OR S412
NO
S603
SW2=ON?
NO
YES
SHOOT BARCODE IMAGE AND TRANSMIT
S604
S605
RECEIVED RESPONSE?
NO
YES
F

FIG. 6B


F
S606
IS RESPONSE NORMAL ?
NO
YES
S607
DISPLAY CONFIRMATION SCREEN
S608
DISPLAY ERROR SCREEN
S609
END?
NO
YES
TO S410 OR S412
S610
RE-SHOOT?
NO
YES
S611
OK?
NO
YES
S612
RE-SHOOT?
NO
YES
G
S613
SET MEDICAL INFORMATION (PATIENT INFORMATION OR PHOTOGRAPHER INFORMATION)
TO S430

F I G. 7A


PLEASE SELECT AREA
HEAD
FACE
NECK
ARM
FEET
MENU
701
702

F I G. 7B

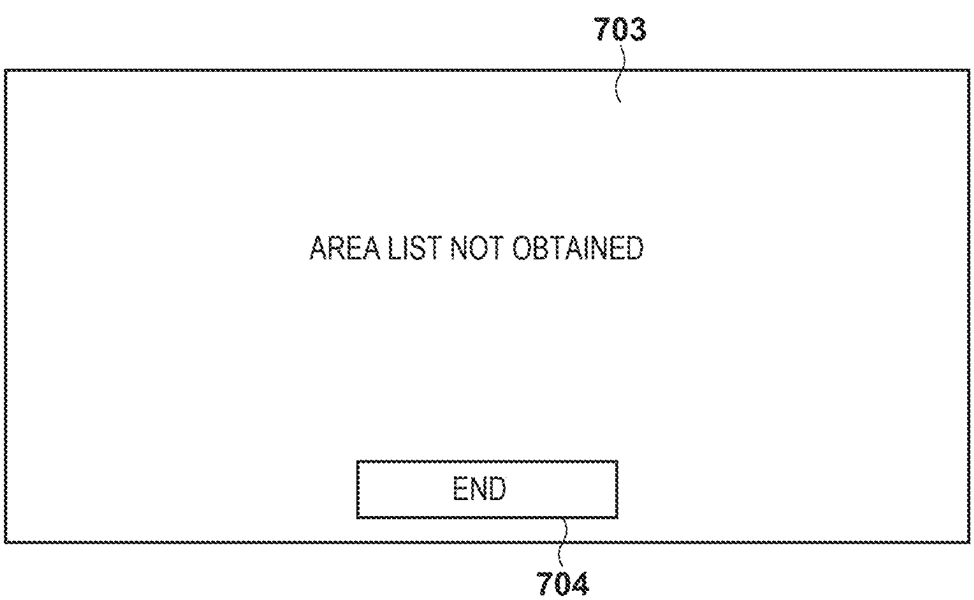

703
AREA LIST NOT OBTAINED
END
704

FIG. 8

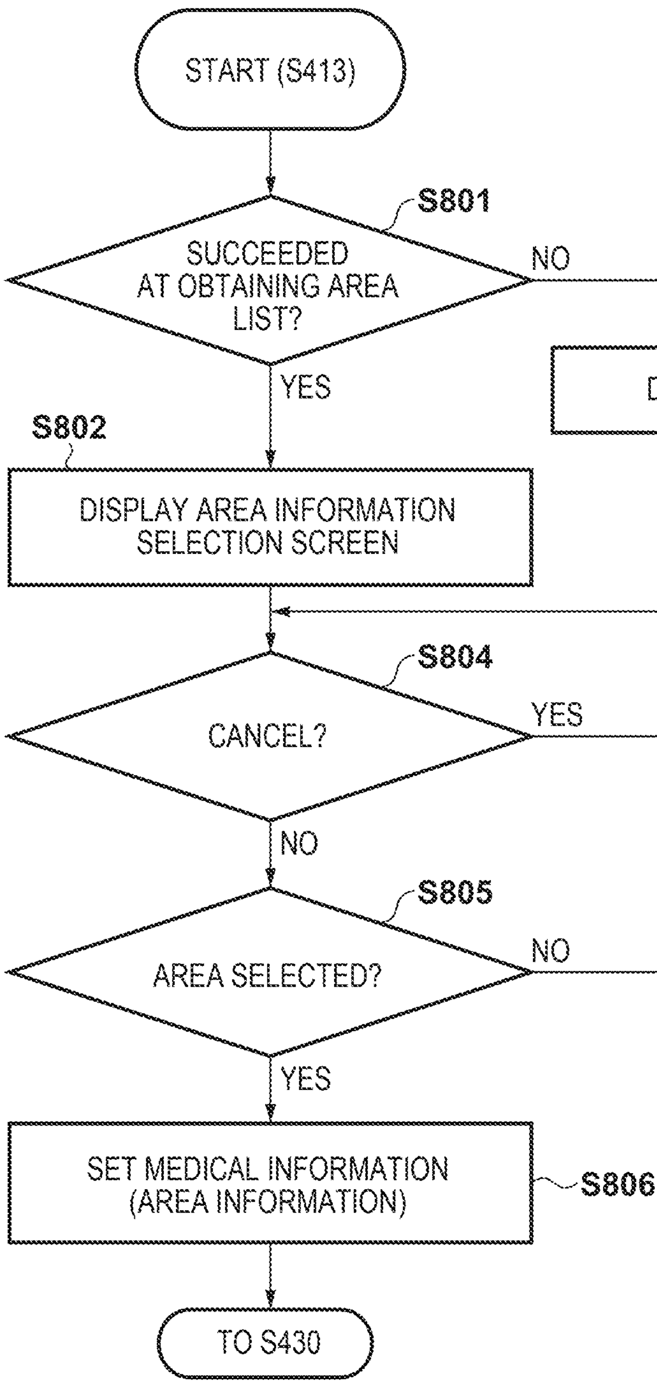

START (S413)
S801
SUCCEEDED AT OBTAINING AREA LIST?
NO
YES
S803
DISPLAY ERROR SCREEN
TO S414
S802
DISPLAY AREA INFORMATION SELECTION SCREEN
S804
CANCEL?
YES
TO S414
NO
S805
AREA SELECTED?
NO
YES
SET MEDICAL INFORMATION (AREA INFORMATION)
S806
TO S430

FIG. 9

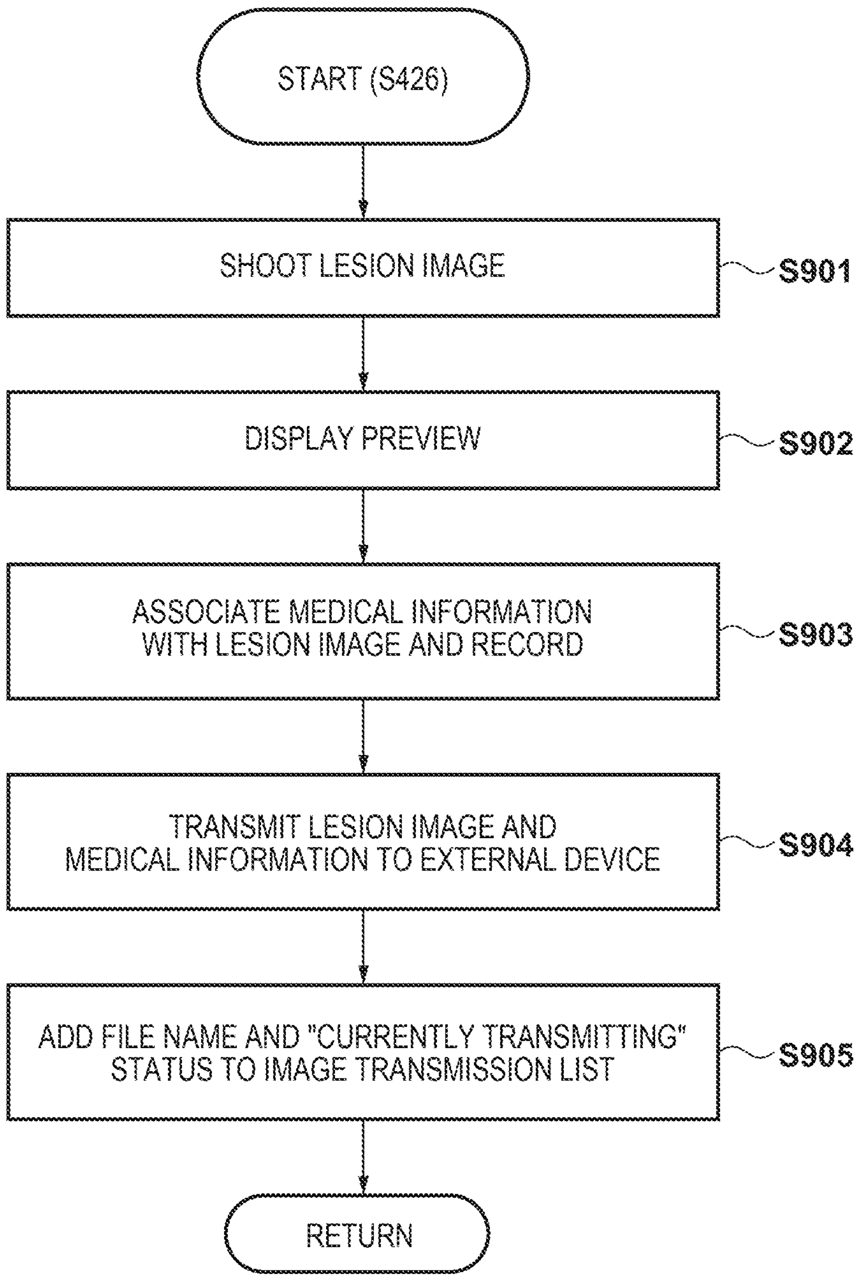

START (S426)
SHOOT LESION IMAGE
S901
DISPLAY PREVIEW
S902
ASSOCIATE MEDICAL INFORMATION WITH LESION IMAGE AND RECORD
S903
TRANSMIT LESION IMAGE AND MEDICAL INFORMATION TO EXTERNAL DEVICE
S904
ADD FILE NAME AND "CURRENTLY TRANSMITTING" STATUS TO IMAGE TRANSMISSION LIST
S905
RETURN

FIG. 10

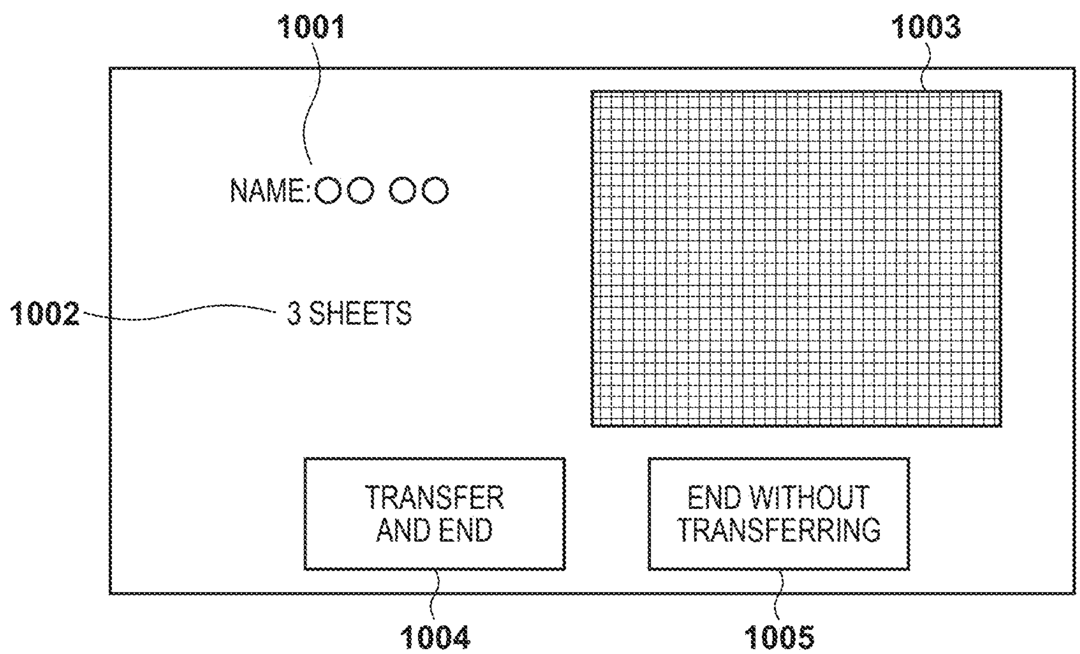
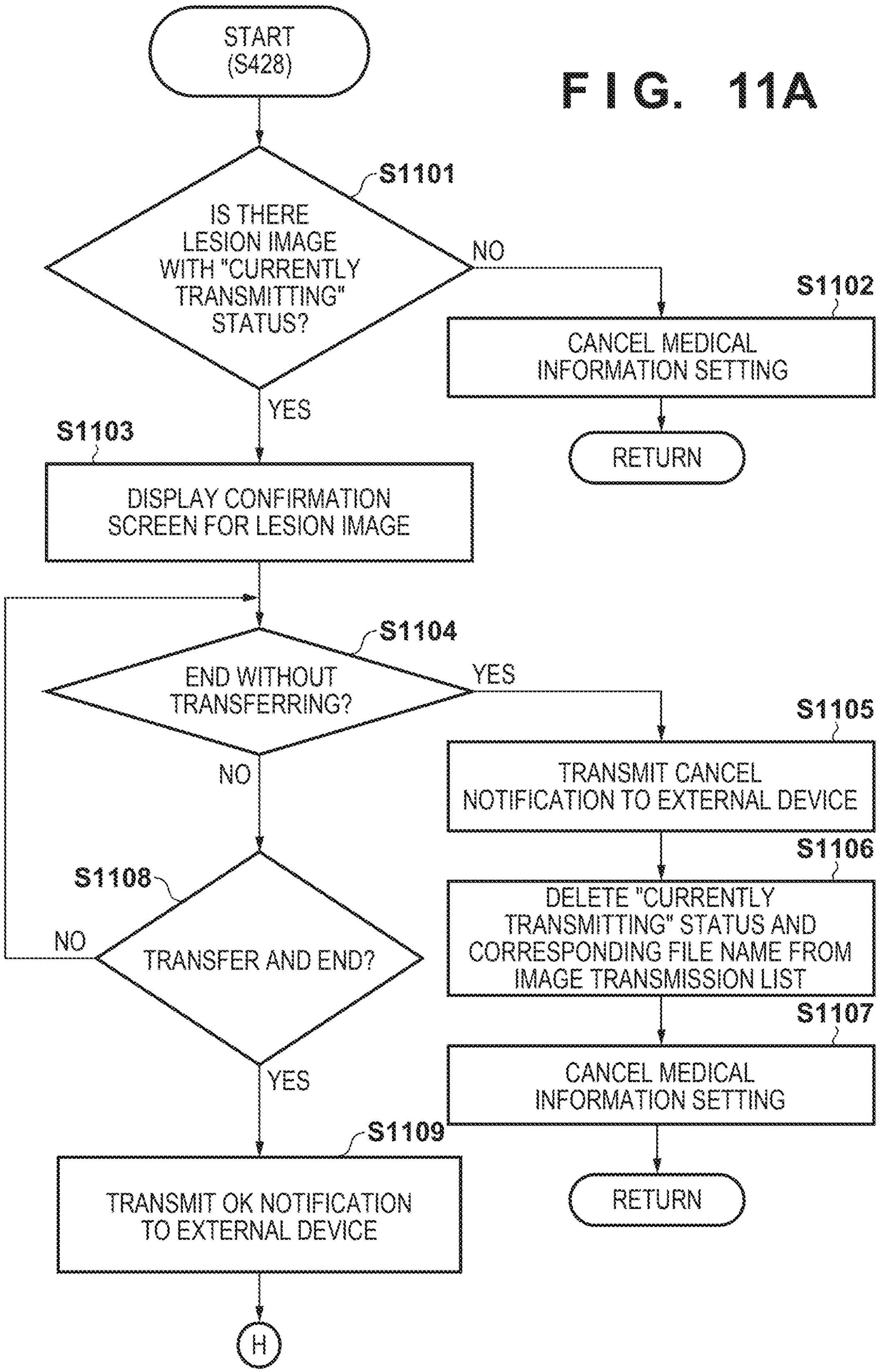

1001
1003
NAME:○○ ○○
1002
3 SHEETS
TRANSFER AND END
END WITHOUT TRANSFERRING
1004
1005

FIG. 11A


START
(S428)
S1101
IS THERE LESION IMAGE WITH "CURRENTLY TRANSMITTING" STATUS?
NO
YES
S1102
CANCEL MEDICAL INFORMATION SETTING
RETURN
S1103
DISPLAY CONFIRMATION SCREEN FOR LESION IMAGE
S1104
END WITHOUT TRANSFERRING?
YES
NO
S1105
TRANSMIT CANCEL NOTIFICATION TO EXTERNAL DEVICE
S1106
DELETE "CURRENTLY TRANSMITTING" STATUS AND CORRESPONDING FILE NAME FROM IMAGE TRANSMISSION LIST
S1107
CANCEL MEDICAL INFORMATION SETTING
RETURN
S1108
TRANSFER AND END?
NO
YES
S1109
TRANSMIT OK NOTIFICATION TO EXTERNAL DEVICE
H

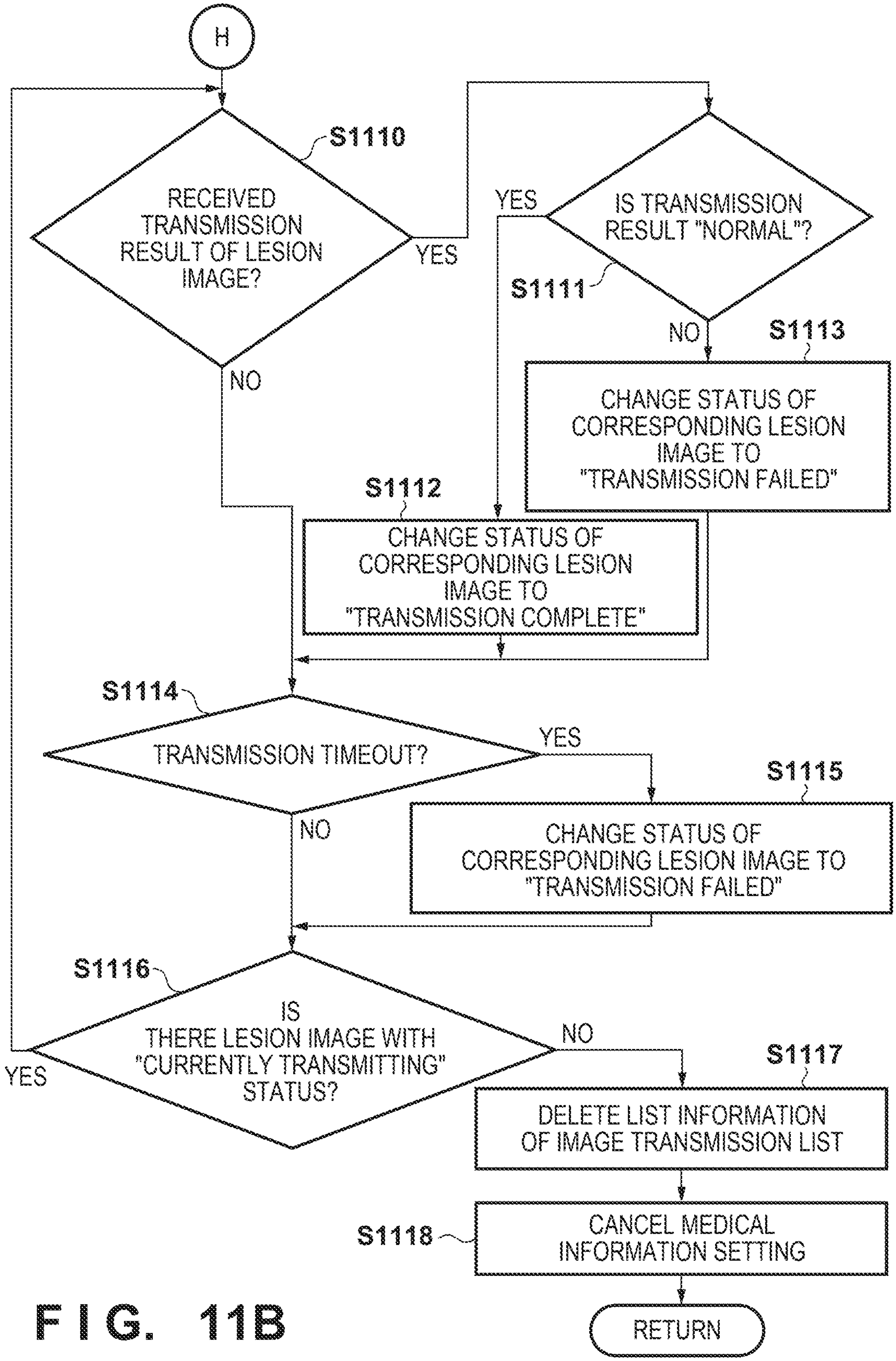

H
S1110
RECEIVED TRANSMISSION RESULT OF LESION IMAGE?
YES
NO
S1111
IS TRANSMISSION RESULT "NORMAL"?
YES
NO
S1113
CHANGE STATUS OF CORRESPONDING LESION IMAGE TO "TRANSMISSION FAILED"
S1112
CHANGE STATUS OF CORRESPONDING LESION IMAGE TO "TRANSMISSION COMPLETE"
S1114
TRANSMISSION TIMEOUT?
YES
NO
S1115
CHANGE STATUS OF CORRESPONDING LESION IMAGE TO "TRANSMISSION FAILED"
S1116
IS THERE LESION IMAGE WITH "CURRENTLY TRANSMITTING" STATUS?
YES
NO
S1117
DELETE LIST INFORMATION OF IMAGE TRANSMISSION LIST
S1118
CANCEL MEDICAL INFORMATION SETTING
RETURN

FIG. 11B

FIG. 12

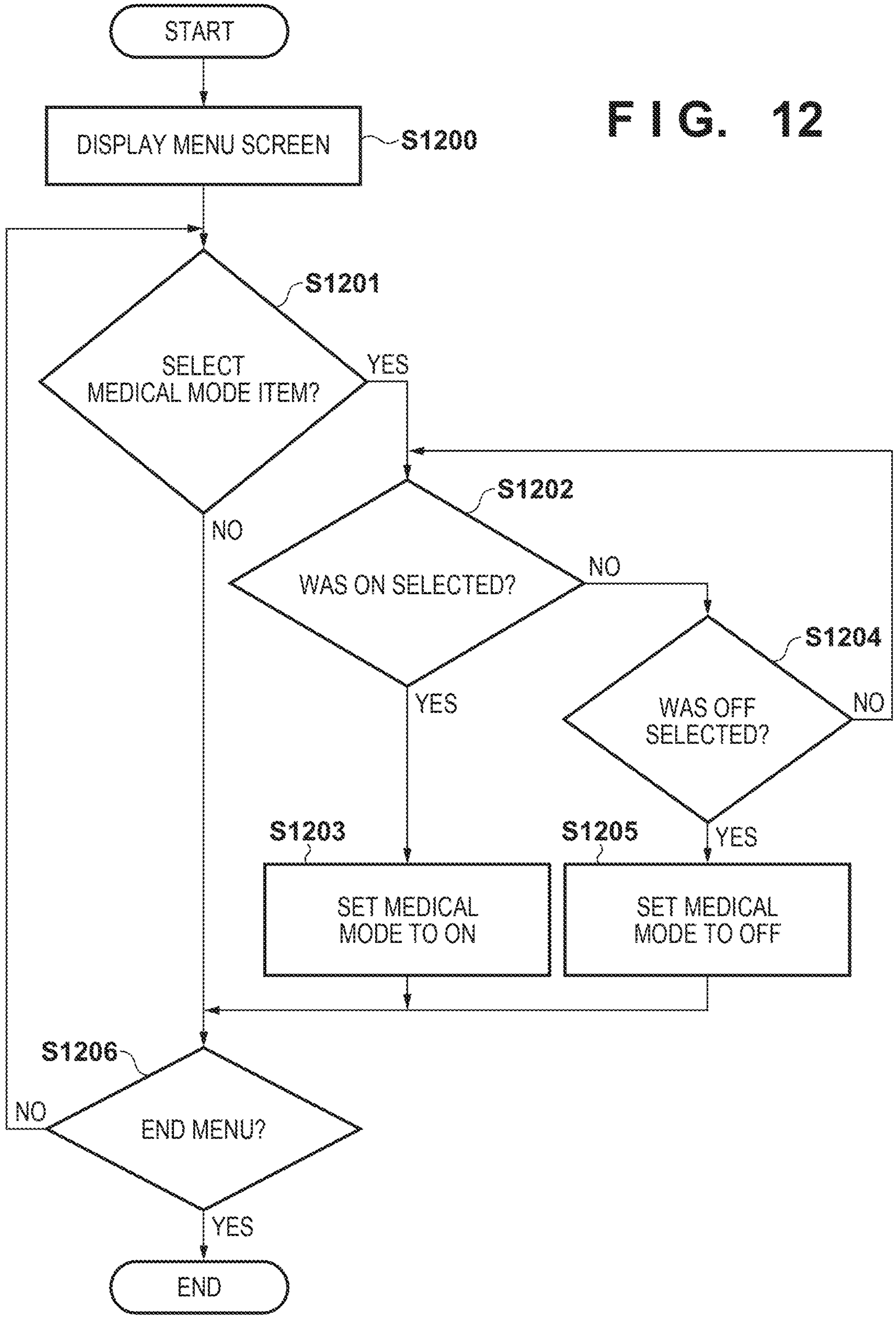

START
DISPLAY MENU SCREEN
S1200
S1201
SELECT MEDICAL MODE ITEM?
YES
NO
S1202
WAS ON SELECTED?
NO
YES
S1204
WAS OFF SELECTED?
NO
YES
S1203
SET MEDICAL MODE TO ON
S1205
SET MEDICAL MODE TO OFF
S1206
END MENU?
NO
YES
END

FIG. 13A

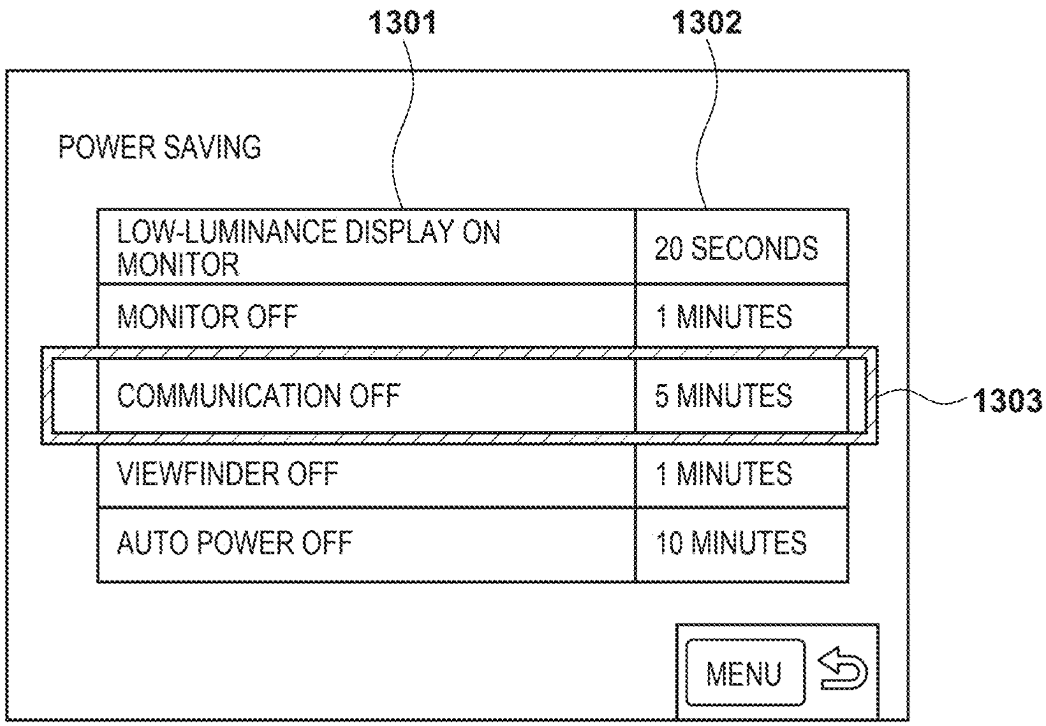


FIG. 13B

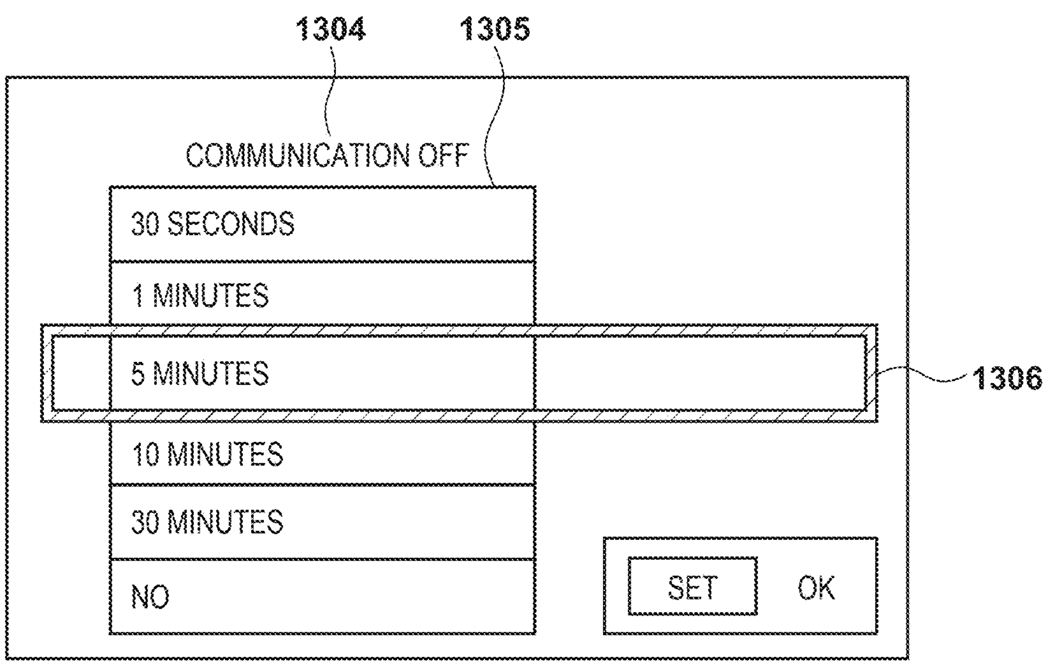


IMAGE CAPTURING APPARATUS, CONTROL METHOD, AND STORAGE MEDIUM

[0001] This application is a Continuation of International Patent Application No. PCT/JP2023/040024, filed Nov. 7, 2023, which claims the benefit of Japanese Patent Application No. 2022-195862, filed Dec. 7, 2022, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

Field of the Technology

[0002] The present disclosure relates to an image capturing apparatus, a control method, and a storage medium.

Description of the Related Art

[0003] A technique is known that records images shot using an image capturing apparatus, such as a digital camera, in association with specific information. For example, in a medical setting, there are cases where shot images are recorded in association with patient information and the like.

[0004] Furthermore, a technique is also known that causes an image capturing apparatus to automatically transition to a power saving state (e.g., a sleep state) in a case where a user has not performed an operation for a certain time period, in order to suppress a power consumption of the image capturing apparatus.

[0005] PTL 1 discloses a technique to cause a tablet terminal to return to a normal state from a sleep state. According to PTL 1, in a system that manages drug administration with use of a tablet terminal installed in a hospital room, the tablet terminal in a sleep state returns to a normal state upon obtaining a patient ID, an injection drug ID, and a nurse ID.

[0006] It takes some time for an image capturing apparatus to return to a normal state from a power saving state. Therefore, if the image capturing apparatus transitions to the power saving state at a timing when a user wishes to perform shooting, shooting cannot be performed promptly. Turning OFF the setting to cause the image capturing apparatus to automatically transition to the power saving state can prevent the image capturing apparatus from transitioning to the power saving state against the user's intention; however, in this case, there is a possibility that a wasteful power consumption increases, and a battery duration is shortened. Conventional techniques, including PTL 1, cannot address such a problem.

CITATION LIST

Patent Literature

[0007] PTL 1: Japanese Patent Laid-Open No. 2020-140219

SUMMARY

[0008] The present disclosure, in at least one aspect thereof, provides a technique to reduce the possibility that an image capturing apparatus transitions to a power saving state against the user's intention while suppressing an increase in a wasteful power consumption.

[0009] According to an aspect of the present disclosure, there is provided an image capturing apparatus including an image capturing unit, comprising: a switching unit configured to switch between ON and OFF of a medical mode of the image capturing apparatus, wherein the medical mode is an operation mode for shooting a medical image; a setting unit configured to set specific information; an association unit configured to associate the set specific information with an image that has been shot using the image capturing unit; and a control unit configured to perform, in a case where the medical mode is ON, control to in a case where the specific information has not been set, cause the image capturing apparatus to transition to a power saving state when a state where a user has not operated the image capturing apparatus has continued for a predetermined time period, and in a case where the specific information has been set, not cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

[0010] Features of the present disclosure will become apparent from the following description of embodiments with reference to the attached drawings. The following description of embodiments are described by way of example.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the disclosure and, together with the description, serve to explain principles of the disclosure.

[0012] FIG. **1**A is an external view of a digital camera **100**.

[0013] FIG. **1**B is an external view of the digital camera **100**.

[0014] FIG. **2** is a block diagram showing an exemplary configuration of the digital camera **100**.

[0015] FIG. **3**A is a diagram showing an example of an LV screen for a case where an LV display setting is set to a medical information display.

[0016] FIG. **3**B is a diagram showing an example of the LV screen for a case where the LV display setting is set to a shooting parameter display.

[0017] FIG. **4**A is a flowchart showing processing executed by the digital camera **100** in a case where a medical mode is ON.

[0018] FIG. **4**B is a flowchart showing processing executed by the digital camera **100** in a case where the medical mode is ON.

[0019] FIG. **4**C is a flowchart showing processing executed by the digital camera **100** in a case where the medical mode is ON.

[0020] FIG. **4**D is a flowchart showing processing executed by the digital camera **100** in a case where the medical mode is ON.

[0021] FIG. **5**A is a diagram showing an example of a barcode obtainment screen displayed in a patient information obtainment mode and a photographer information obtainment mode.

[0022] FIG. **5**B is a diagram showing an example of a confirmation screen of patient information (in the case of the patient information obtainment mode) or photographer information (in the case of the photographer information obtainment mode) corresponding to a shot barcode.

[0023] FIG. **5**C is a diagram showing an example of an error screen displayed in a case where the obtainment (reception) of the patient information or the photographer information has failed.

[0024] FIG. **6**A is a flowchart of processing in the patient information obtainment mode and the photographer information obtainment mode.

[0025] FIG. **6**B is a flowchart of processing in the patient information obtainment mode and the photographer information obtainment mode.

[0026] FIG. **7**A is a diagram showing an example of an area information selection screen.

[0027] FIG. **7**B is a diagram showing an example of an error screen.

[0028] FIG. **8** is a flowchart of processing in an area information obtainment mode.

[0029] FIG. **9** is a flowchart of processing in a lesion shooting mode.

[0030] FIG. **10** is a diagram showing an example of a confirmation screen for a lesion image.

[0031] FIG. **11**A is a flowchart of processing in a lesion shooting end mode.

[0032] FIG. **11**B is a flowchart of processing in the lesion shooting end mode.

[0033] FIG. **12** is a flowchart of processing for switching between ON and OFF of the medical mode.

[0034] FIG. **13**A is a diagram showing an example of a list display screen of power saving items.

[0035] FIG. **13**B is a diagram showing an example of a detailed setting screen of a power saving item corresponding to “communication OFF”.

DESCRIPTION OF THE EMBODIMENTS

[0036] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claims. Multiple features are described in the embodiments, but it is not the case that all such features are required, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

External View of Digital Camera **100**

[0037] FIG. **1**A and FIG. **1**B are external views of a digital camera **100**, which is one example of an image capturing apparatus. FIG. **1**A is a front perspective view of the digital camera **100**, and FIG. **1**B is a rear perspective view of the digital camera **100**.

[0038] A display unit **28** is a display unit provided on a rear surface of the digital camera **100**, and displays images and various types of information. A touch panel **70***a* can detect a touch operation on a display surface (a touch operation surface) of the display unit **28**. An out-of-view-finder display unit **43** is a display unit provided on a top surface of the digital camera **100**, and displays a variety of setting values of the digital camera **100**, including a shutter speed and a diaphragm. A shutter button **61** is an operation member for issuing a shooting instruction. A mode changing switch **60** is an operation member for changing among various types of modes. Terminal covers **40** are covers that protect connectors (not shown) for connecting connection cables intended to connect the digital camera **100** to an external device.

[0039] A main electronic dial **71** is a rotary operation member, and rotating the main electronic dial **71** can, for example, change such setting values as the shutter speed and the diaphragm. A power source switch **72** is an operation member that switches between ON and OFF of a power source of the digital camera **100**. A sub electronic dial **73** is a rotary operation member, and rotating the sub electronic dial **73** enables, for example, moving of a selection frame (cursor) and an image jump. A four-direction key **74** is configured in such a manner that upper, lower, left, and right portions thereof can each be pushed. The digital camera **100** executes processing corresponding to a pushed portion of the four-direction key **74**. A SET button **75** is a push button, and is mainly used in, for example, deciding on a selected item.

[0040] A moving image button **76** is used to issue an instruction for starting and stopping the shooting (recording) of moving images. An AE lock button **77** is a push button, and depressing the AE lock button **77** in a shooting standby state can fix an exposure state. A magnification button **78** is an operation button for switching between ON and OFF of a magnification mode during live-view display (LV display) in a shooting mode. A live-view image (LV image) can be magnified or reduced by operating the main electronic dial **71** while the magnification mode is ON. In a reproduction mode, the magnification button **78** functions as an operation button for magnifying a reproduced image and increasing the magnification rate thereof. A reproduction button **79** is an operation button for switching between the shooting mode and the reproduction mode. Depressing the reproduction button **79** during the shooting mode causes the digital camera **100** to transition to the reproduction mode; as a result, among the images recorded in a recording medium **200** (described later), the newest image can be displayed on the display unit **28**. A menu button **81** is a push button that is used to perform an instruction operation for displaying a menu screen; when the menu button **81** has been pushed, the display unit **28** displays the menu screen on which various types of settings can be configured. A user can intuitively configure various types of settings with use of the menu screen displayed on the display unit **28**, the four- direction key **74**, and the SET button **75**. A display switching button **83** is used to switch a display setting of the display unit **28**.

[0041] A touch bar **82** (multi-function bar/M-Fn bar) is a line-like touch operation member (line touch sensor) that can accept a touch operation. The touch bar **82** is located at a position where it can be touch-operated (can be touched) by a thumb of a right hand in a state where a grip unit **90** is grasped by the right hand (in a state where it is grasped by a little finger, a ring finger, and a middle finger of the right hand) in such a manner that the shutter button **61** can be depressed by a pointing finger of the right hand. That is to say, the touch bar **82** is located at a position where it can be operated in a state where an eye is looking through a viewfinder while in proximity to an eye proximity unit **16** and the digital camera is held in such a manner that the shutter button **61** can be depressed at any time (a shooting posture). The touch bar **82** is an acceptance unit that can accept a tap operation (an operation of touching and then releasing within a predetermined time period without moving a touch position), a leftward/rightward slide operation (an operation of touching and then moving a touch position while maintaining the touch), and the like on the touch bar **82**. The touch bar **82** is an operation member different from the touch panel **70***a*, and does not have a display function.

[0042] A communication terminal **10** is a communication terminal intended for the digital camera **100** to communicate with an attachable/removable lens unit **150** (described later). The eye proximity unit **16** is an eye proximity unit for an eye proximity viewfinder **17** (look-through viewfinder), and the user can visually recognize a video displayed on an internal EVF **29** (Electronic View Finder) via the eye proximity unit **16**. An eye proximity detection unit **57** is an eye proximity detection sensor that detects whether an eye of the user (photographer) is in proximity to the eye proximity unit **16**. A cover **202** is a cover for a slot in which the recording medium **200** (described later) is stored. The grip unit **90** is a holding unit with a shape that is easily grasped by the right hand while the user is holding the digital camera **100**. The shutter button **61** and the main electronic dial **71** are located at positions where they can be operated by the pointing finger of the right hand in a state where the digital camera **100** is held while the grip unit **90** is grasped by the little finger, the ring finger, and the middle finger of the right hand. Furthermore, the sub electronic dial **73** and the touch bar **82** are located at positions where they can be operated by the thumb of the right hand in the same state. A thumb rest unit **91** (thumb standby position) is a grip member provided at a location which is on the rear surface side of the digital camera **100**, and at which the thumb of the right hand grasping the grip unit **90** easily rests in a state where the thumb is not operating any operation member. The thumb rest unit **91** is made up of, for example, a rubber member for increasing a holding force (grip feeling).

Block Diagram of Digital Camera **100**

[0043] FIG. **2** is a block diagram showing an exemplary configuration of the digital camera **100**. The lens unit **150** is an interchangeable lens unit provided with a shooting lens (lens **103**). The lens **103** is normally composed of a plurality of lenses, but is represented as only one lens in FIG. **2** for simplicity.

[0044] A communication terminal **6** is a communication terminal intended for the lens unit **150** to communicate with the digital camera **100** side, and a communication terminal **10** is a communication terminal intended for the digital camera **100** to communicate with the lens unit **150** side. The lens unit **150** communicates with a system control unit **50** via these communication terminals **6** and **10**. Also, the lens unit **150** causes an internal lens system control circuit **4** to control a diaphragm **1** via a diaphragm driving circuit **2**. Furthermore, the lens unit **150** achieves focus by causing the lens system control circuit **4** to displace the position of the lens **103** via an AF driving circuit **3**.

[0045] A shutter **101** is a focal-plane shutter that can freely control an exposure time period of an image capturing unit **22** under control of the system control unit **50**.

[0046] The image capturing unit **22** is an image capturing element (image sensor) composed of, for example, a CCD or CMOS element that converts an optical image into electrical signals. The image capturing unit **22** may include an image capturing plane phase-difference sensor that outputs defocus amount information to the system control unit **50**. An A/D converter **23** converts analog signals output from the image capturing unit **22** into digital signals.

[0047] An image processing unit **24** executes predetermined processing (e.g., pixel interpolation processing, resize processing like reduction, and color conversion processing) with respect to data from the A/D converter **23**, or data from a memory control unit **15**. Also, The image processing unit **24** executes predetermined computation processing with use of captured image data. The system control unit **50** performs exposure control and range-finding control on the basis of the computation result obtained by the image processing unit **24**. As a result, for example, autofocus (AF) processing, automatic exposure (AE) processing, and preliminary flash emission (EF) processing of the through-the-lens (TTL) method are executed. The image processing unit **24** further executes predetermined computation processing with use of captured image data, and executes auto white balance (AWB) processing of the TTL method on the basis of the obtained computation result.

[0048] Output data from the A/D converter **23** is written into a memory **32** via the image processing unit **24** and the memory control unit **15**. Alternatively, output data from the A/D converter **23** is written into the memory **32** via the memory control unit **15** without intervention of the image processing unit **24**. The memory **32** stores image data that has been obtained by the image capturing unit **22** and converted into digital data by the A/D converter **23**, and image data to be displayed on the display unit **28** and the EVF **29**. The memory **32** has a storage capacity sufficient to store a predetermined number of still images, and moving images and audio of a predetermined duration.

[0049] Furthermore, the memory **32** also functions as a memory for image display (video memory). A D/A converter **19** converts image data for display stored in the memory **32** into analog signals, and supplies the analog signals to the display unit **28** and the EVF **29**. In this way, image data for display that has been written into the memory **32** is displayed by the display unit **28** and the EVF **29** via the D/A converter **19**. Each of the display unit **28** and the EVF **29** is a display, such as an LCD and organic EL, and performs display in accordance with analog signals from the D/A converter **19**. Digital signals that have undergone A/D conversion in the A/D converter **23** and have been accumulated in the memory **32** are converted into analog signals by the D/A converter **19**, and sequentially transferred to and displayed by the display unit **28** or the EVF **29**; as a result, live-view display (LV display) can be performed. Hereinafter, images that are displayed in the form of live-view display will be referred to as live-view images (LV images).

[0050] The system control unit **50** is a control unit composed of at least one processor and/or at least one circuit, and controls an entirety of the digital camera **100**. The system control unit **50** realizes each process of the present embodiment, which will be described later, by executing a program recorded in a nonvolatile memory **56**. Furthermore, the system control unit **50** also performs display control by controlling the memory **32**, the D/A converter **19**, the display unit **28**, the EVF **29**, and so forth.

[0051] A system memory **52** is, for example, a RAM. The system control unit **50** deploys constants and variables for the operations of the system control unit **50**, the program that has been read out from the nonvolatile memory **56**, and the like to the system memory **52**.

[0052] The nonvolatile memory **56** is an electrically erasable and recordable memory, and is, for example, an EEPROM and the like. The constants for the operations of the system control unit **50**, the program, and the like are recorded in the nonvolatile memory **56**. The program mentioned here denotes a program for executing processing of various types of flowcharts, which will be described later in the present embodiment.

[0053] A system timer **53** is a time measurement unit that measures the time periods used in various types of control, and the time of a built-in clock.

[0054] A communication unit **54** transmits/receives video signals and audio signals to/from an external device that is connected wirelessly or via a wired cable. The communication unit **54** is also connectable to a wireless Local Area Network (wireless LAN) and the Internet. Furthermore, the communication unit **54** can communicate with an external device also via Bluetooth® and Bluetooth Low Energy. The communication unit **54** can transmit images captured by the image capturing unit **22** (including LV images) and images recorded in the recording medium **200**, and can receive image data and other various types of information from an external device.

[0055] An attitude detection unit **55** detects an attitude of the digital camera **100** relative to the gravity direction. Based on the attitude detected by the attitude detection unit **55**, it is possible to distinguish whether an image shot by the image capturing unit **22** is an image that was shot while the digital camera **100** was held in a landscape orientation or an image that was shot while the digital camera **100** was held in a portrait orientation. The system control unit **50** can add direction information corresponding to the attitude detected by the attitude detection unit **55** to an image file of an image captured by the image capturing unit **22**. Furthermore, the system control unit **50** can rotate an image and record the resultant image on the basis of direction information corresponding to the attitude detected by the attitude detection unit **55**. For example, an acceleration sensor or a gyro sensor can be used as the attitude detection unit **55**. A motion of the digital camera **100** (e.g., whether the digital camera **100** is panned, tilted, lifted, or stationary) can also be detected using the acceleration sensor or the gyro sensor acting as the attitude detection unit **55**.

[0056] The eye proximity detection unit **57** is an eye proximity detection sensor that detects approaching (eye proximity) and separation (eye separation) of an eye (object) to and from and the eye proximity unit **16** of the eye proximity viewfinder **17** (approach detection). The system control unit **50** switches between display (a displaying state) and non-display (a non-displaying state) of the display unit **28** and the EVF **29** in accordance with the state detected by the eye proximity detection unit **57**. More specifically, for example, in a case where the setting of switching of a display destination is automatic switching during the shooting standby state, the system control unit **50** sets the display unit **28** as the display destination and turns ON its display, and causes the EVF **29** to display nothing, while the eye is not in proximity. Also, the system control unit **50** sets the EVF **29** as the display destination and turns ON its display, and causes the display unit **28** to display nothing, while the eye is in proximity.

[0057] For example, an infrared proximity sensor can be used as the eye proximity detection unit **57**, and approaching of some sort of object to the eye proximity unit **16** of the eye proximity viewfinder **17**, in which the EVF **29** is built, can be detected. In a case where an object has approached, infrared light projected from a light projection unit (not shown) of the eye proximity detection unit **57** is reflected by the object and received by a light receiving unit (not shown) of the infrared proximity sensor. It is also possible to distinguish at what distance from the eye proximity unit **16** the approaching object is situated (an eye proximity distance) on the basis of the amount of the received infrared light. In this way, the eye proximity detection unit **57** performs eye proximity detection for detecting a distance at which the object is approaching the eye proximity unit **16**. It is assumed that eye proximity is detected in a case where an approaching object at a predetermined distance or less from the eye proximity unit **16** has been detected in a non-eye proximity state (non-approaching state). It is assumed that eye separation is detected in a case where an object whose approach was detected has separated by a predetermined distance or more in an eye proximity state (approaching state). A threshold for detection of eye proximity and a threshold for detection of eye separation may be different from each other due to, for example, provision of hysteresis and the like. Furthermore, it is assumed that a detection of eye proximity is followed by the eye proximity state, until eye separation is detected. It is assumed that a detection of eye separation is followed by the non-eye proximity state, until eye proximity is detected. Note that the infrared proximity sensor is one example, and another sensor may be used as the eye proximity detection unit **57** as long as it can detect a state that can be regarded as eye proximity.

[0058] The out-of-viewfinder display unit **43** displays a variety of setting values of the camera, including the shutter speed and the diaphragm, via an out-of-viewfinder display unit driving circuit **44**.

[0059] A power source control unit **80** is composed of a battery detection circuit, a DC-DC converter, a switch circuit that changes a block to which current is supplied, and the like, and detects whether a battery is loaded, a battery type, and a remaining battery level, for example. Also, the power source control unit **80** controls the DC-DC converter on the basis of the results of the foregoing detection and an instruction from the system control unit **50**, and supplies a necessary voltage for a necessary time period to each unit, including the recording medium **200**. A power source unit **30** is composed of, for example, a primary battery like an alkaline battery or a lithium battery, a secondary battery like a NiCd battery, a NiMH battery, or a Li battery, or an AC adapter.

[0060] A recording medium I/F **18** is an interface with the recording medium **200**, which is a memory card, a hard disk, and the like. The recording medium **200** is a recording medium, such as a memory card, for recording shot images, and is composed of a semiconductor memory, a magnetic disk, and the like.

[0061] An operation unit **70** is an input unit that accepts an operation from the user (a user operation), and is used to input various types of operational instructions to the system control unit **50**. As shown in FIG. **2**, the operation unit **70** includes the shutter button **61**, the mode changing switch **60**, the power source switch **72**, the touch panel **70***a*, and other operation members **70***b*. Other operation members **70***b* include, for example, the main electronic dial **71**, the sub electronic dial **73**, the four-direction key **74**, the SET button **75**, the moving image button **76**, the AE lock button **77**, the magnification button **78**, the reproduction button **79**, the menu button **81**, and the touch bar **82**.

[0062] The shutter button **61** includes a first shutter switch **62** and a second shutter switch **64**. The first shutter switch **62** is turned ON and generates a first shutter switch signal SW**1** halfway through an operation performed on the shutter button **61**, that is to say, when the shutter button **61** is half-pressed (a shooting preparation instruction). In response to the first shutter switch signal SW**1**, the system control unit **50** starts shooting preparation operations, such as autofocus (AF) processing, automatic exposure (AE) processing, auto white balance (AWB) processing, and preliminary flash emission (EF) processing.

[0063] The second shutter switch **64** is turned ON and generates a second shutter switch signal SW**2** upon completion of the operation performed on the shutter button **61**, that is to say, when the shutter button **61** is full-pressed (a shooting instruction). In response to the second shutter switch signal SW**2**, the system control unit **50** starts a sequence of operations of shooting processing, from readout of signals from the image capturing unit **22** to writing of a captured image as an image file into the recording medium **200**.

[0064] The mode changing switch **60** changes an operation mode of the system control unit **50** to one of a still image shooting mode, a moving image shooting mode, a reproduction mode, and so forth. Modes included in the still image shooting mode include an auto shooting mode, an auto scene distinction mode, a manual mode, a diaphragm priority mode (Av mode), a shutter speed priority mode (Tv mode), and a program AE mode (P mode). They also include various types of scene modes in which shooting settings vary for each shooting scene, a custom mode, and so forth. The mode changing switch **60** enables the user to change directly to one of these modes. Alternatively, it is permissible to adopt a configuration in which, after the user has changed to a screen of a list of shooting modes with use of the mode changing switch **60**, the user can selectively change to one of the displayed plurality of modes with use of another operation member. Similarly, the moving image shooting mode may also include a plurality of modes.

[0065] The touch panel **70***a* is a touch sensor that detects various types of touch operations on the display surface of the display unit **28** (an operation surface of the touch panel **70***a*). The touch panel **70***a* and the display unit **28** can be configured integrally. For example, the touch panel **70***a* is configured so that a light transmittance thereof does not interfere with display of the display unit **28**, and is attached to a top layer of the display surface of the display unit **28**. Also, input coordinates on the touch panel **70***a* are associated with display coordinates on the display surface of the display unit **28**. This makes it possible to provide a graphical user interface (GUI) via which the user can operate a screen displayed on the display unit **28**, as if directly.

[0066] The system control unit **50** can detect the following operations or states with respect to the touch panel **70***a*.

- **[0067]** Newly touching the touch panel **70***a* with a finger or a stylus that was not touching the touch panel **70***a*, that is to say, the start of a touch (hereinafter referred to as "touch-down")
- **[0068]** A state where a finger or a stylus is touching the touch panel **70***a* (hereinafter referred to as "touch-on")
- **[0069]** Movement of a finger or a stylus while it is touching the touch panel **70***a* (hereinafter referred to as "touch-move")
- **[0070]** Separation (releasing) of a finger or a stylus that was touching the touch panel **70***a* from the touch panel **70***a*, that is to say, the end of a touch (hereinafter referred to as "touch-up")
- **[0071]** A state where nothing is touching the touch panel **70***a* (hereinafter referred to as "touch-off")

[0072] When a touch-down is detected, a touch-on is detected at the same time. After a touch-down, a touch-on normally continues to be detected as long as no touch-up is detected. Also in a case where a touch-move is detected, a touch-on is detected at the same time. Even when a touch-on is detected, a touch-move is not detected as long as a touch position does not move. A touch-off follows the detection of a touch-up of every finger or stylus that was touching.

[0073] The system control unit **50** is notified of these operations and states, as well as positional coordinates at which a finger or a stylus is touching the touch panel **70***a*, via an internal bus. Then, the system control unit **50** judges what kind of operation (touch operation) has been performed on the touch panel **70***a* on the basis of information it has been notified of. Regarding a touch-move, a moving direction of a finger or a stylus moving on the touch panel **70***a* can also be judged for each of a vertical component and a horizontal component on the touch panel **70***a* on the basis of a change in positional coordinates. It is assumed that, in a case where a touch-move of a predetermined distance or longer has been detected, it is judged that a slide operation has been performed. An operation of quickly moving a finger by some distance while the finger is touching the touch panel **70***a* and then releasing the finger is called a flick. In other words, a flick is a quick tracing operation involving a swipe of a finger on the touch panel **70***a*. It can be judged that a flick has been performed when a touch-up is detected immediately after the detection of a touch-move of a predetermined distance or longer at a predetermined speed or faster (it can be judged that a flick has been performed following a slide operation). Furthermore, a touch operation of bringing the touch positions close to each other and a touch operation of moving the touch positions away from each other, while touching a plurality of locations (e.g., two points) at the same time (performing a multi-touch), are referred to as a pinch-in and a pinch-out, respectively. A pinch-out and a pinch-in are collectively referred to as a pinch operation (or simply a pinch). The touch panel **70***a* may be of any type included among a variety of types of touch panels, such as a resistive film type, a capacitance type, a surface acoustic wave type, an infrared type, an electromagnetic induction type, an image recognition type, and an optical sensor type. There are a type in which a touch is detected on the occurrence of contact with the touch panel, and a type in which a touch is detected when a finger or a stylus has approached the touch panel; either type may be used.

Basic Operations of Digital Camera **100**

[0074] Basic operations of the digital camera **100** according to the present embodiment will be described. In the following description, it is assumed that the digital camera **100** is used to shoot a lesion image, which is one example of a medical image. Also, it is assumed that the digital camera **100** is configured to associate medical information with a lesion image. Medical information includes, for example, at least one of patient information (information related to a patient), photographer information (information related to a photographer), and area information (information related to a shooting target area of a patient). However, images shot by the digital camera **100** are not limited to lesion images, and information that the digital camera **100** associates with lesion images is not limited to medical information. The technique ideas of the present embodiment are applicable to an image capturing apparatus that associates specific information with shot images, regardless of the types of images and information.

[0075] The digital camera **100** is configured to communicate with an external system including an external device via the communication unit **54**. In the following description, it is assumed that the external system is a medical system. The digital camera **100** is configured to be capable of switching between ON (enabled) and OFF (disabled) of a medical mode (a predetermined operation mode). The digital camera **100** includes a patient information obtainment mode, a photographer information obtainment mode, an area information obtainment mode, a lesion shooting mode, and a lesion shooting end mode, as operation modes in a state where the medical mode is ON.

[0076] In the patient information obtainment mode, the digital camera **100** transmits image data including a code (e.g., a one-dimensional code or a two-dimensional code), which has been obtained through shooting or an LV operation, to the external device via the communication unit **54**. The digital camera **100** receives patient information (e.g., an ID, a name, a sex, an age, and the like) corresponding to the code included in the transmitted image data from the external device via the communication unit **54**, and stores the patient information into the system memory **52**. Note that the digital camera **100** may obtain code information by analyzing the code included in the image data, and transmit the code information, in place of the image data, to the external device. In the following description, it is assumed that image data including a barcode (a barcode image) is obtained as the image data including the code in the patient information obtainment mode.

[0077] In the photographer information obtainment mode, the digital camera **100** transmits image data including a code (e.g., a one-dimensional code or a two-dimensional code), which has been obtained through shooting or an LV operation, to the external device via the communication unit **54**. The digital camera **100** receives photographer information (e.g., an ID, a name, a clinical department, and the like) corresponding to the code included in the transmitted image data from the external device via the communication unit **54**, and stores the photographer information into the system memory **52**. Note that the digital camera **100** may obtain code information by analyzing the code included in the image data, and transmit the code information, in place of the image data, to the external device. In the following description, it is assumed that image data including a barcode (a barcode image) is obtained as the image data including the code in the photographer information obtainment mode.

[0078] In the area information obtainment mode, the digital camera **100** obtains area list information (an area list) from the external device via the communication unit **54**, and stores the area list information into the system memory **52**. Once the user has selected and decided on an area corresponding to a lesion from the area list, the digital camera **100** stores information indicating the selected area (area information) into the system memory **52**.

[0079] In the lesion shooting mode, the digital camera **100** obtains image data of a subject (a lesion of a patient) in accordance with a shooting operation of the user. The digital camera **100** links the obtained image data with the medical information including patient information, photographer information, and area information stored in the system memory **52**, and saves the same in the recording medium **200**. Also, the digital camera **100** transmits the image data and the medical information linked therewith from the communication unit **54** to the external device. A method of linking the image data with the medical information is not limited in particular. For example, the digital camera **100** can embed the medical information in a part of the image data. Alternatively, the digital camera **100** may save the medical information in the recording medium **200** as a file different from a file of the image data, and link the file of the medical information with the file of the image data.

[0080] In the lesion shooting end mode, the digital camera **100** displays thumbnail images of the image data shot in the lesion shooting mode, and the medical information linked with the image data. The user confirms the displayed thumbnail images and medical information, and selects whether to transfer the image data and the medical information to the medical system. Also, the digital camera **100** clears the medical information including the patient information, the photographer information, and the area information held in the system memory **52**, thereby cancelling the setting of the medical information.

Examples of Live-View Screen (LV Screen)

[0081] FIG. **3**A and FIG. **3**B are diagrams showing examples of an LV screen (a shooting standby screen) displayed during a shooting standby in a case where the medical mode is ON. The digital camera **100** displays the LV screen in accordance with an LV display setting. The LV display setting includes a medical information display and a shooting parameter display. The user can switch between the medical information display and the shooting parameter display by depressing the display switching button **83**. Also, the digital camera **100** may switch the LV display setting to the medical information display when the user has half-pressed the shutter button **61**.

[0082] FIG. **3**A is a diagram showing an example of the LV screen for a case where the LV display setting is set to the medical information display. **301** is a display region of the LV image. Icons, text boxes, and buttons indicated by **302** to **309** are displayed while being superimposed on the LV image.

[0083] **302** is an icon indicating a photographer, and indicates that a text box (a photographer information frame **303**) next to it displays information of the photographer. After photographer information has been obtained in the photographer information obtainment mode, the digital camera **100** may switch the icon **302** to an icon with a design tailored to the photographer information.

[0084] **303** is the photographer information frame. When the user has touched the photographer information frame **303** (or when the user has moved a selection frame to the photographer information frame **303** through a dial operation and depressed the SET button **75**), the digital camera **100** transitions to the photographer information obtainment mode. The digital camera **100** displays photographer information (including a photographer name) obtained in the photographer information obtainment mode in the photographer information frame **303**.

[0085] Note that in the present specification, in a case where some sort of processing is executed through the "dial operation", the "dial operation" refers to an operation on a dial to which this processing has been assigned among the main electronic dial **71** and the sub electronic dial **73** included in the digital camera **100**. For example, in a case where processing for moving the selection frame on the LV screen corresponding to the medical information display is assigned to the sub electronic dial **73**, the "dial operation" in "the user has moved the selection frame to the photographer information frame **303** through the dial operation" corresponds to an operation on the sub electronic dial **73**.

[0086] **304** is an icon indicating a patient, and indicates that a text box (a patient information frame **305**) next to it displays information of the patient. After patient information has been obtained in the patient information obtainment mode, the digital camera **100** may change the icon **304** to an icon with a design tailored to the patient information. As one example, the icon may be switched in accordance with the patient's sex and age.

[0087] **305** is the patient information frame. When the user has touched the patient information frame **305** (or when the user has moved the selection frame to the patient information frame **305** through a dial operation and depressed the SET button **75**), the digital camera **100** transitions to the patient information obtainment mode. The digital camera **100** displays the patient information (including a patient name) obtained in the patient information obtainment mode in the patient information frame **305**. As one example, the digital camera **100** may display a patient name, a sex, and an age.

[0088] **306** is an icon indicating an area, and indicates that a text box (an area information frame **307**) next to it displays information on area selection. After area information has been obtained in the area information obtainment mode, the digital camera **100** may switch the icon **306** to an icon with a design tailored to the area information.

[0089] **307** is the area information frame. When the user has touched the area information frame **307** (or when the user has moved the selection frame to the area information frame **307** through a dial operation and depressed the SET button **75**), the digital camera **100** transitions to the area information obtainment mode. The digital camera **100** displays area information (including an area name) obtained in the area information obtainment mode in the area information frame **307**.

[0090] When the user has full-pressed the shutter button **61**, the digital camera **100** transitions to the lesion shooting mode. In the lesion shooting mode, the digital camera **100** performs shooting, associates shot image data with medical information, and transmits the same to the external device.

[0091] **308** is an end button. The user depresses the end button **308** when shooting of a necessary number of images has ended. When the end button **308** has been depressed, the digital camera **100** transitions to the lesion shooting end mode. The user selects whether to transfer lesion images to the medical system. In a case where the transfer of the lesion images to the medical system has been selected, the digital camera **100** transfers the lesion images to the medical system. Also, the digital camera **100** clears the photographer information, the patient information, and the area information.

[0092] **309** is an icon indicating a communication state. The icon **309** indicates whether the digital camera **100** is unconnected to the external device or is currently connected to the external device.

[0093] FIG. **3**B is a diagram showing an example of the LV screen for a case where the LV display setting is set to the shooting parameter display.

[0094] **311** is a display region of the LV image. **312** indicates a shooting mode. **313** indicates a setting value of the shutter speed. **314** indicates an f-number. **315** indicates an exposure correction value. **316** indicates an ISO value.

Processing in Medical Mode

[0095] FIGS. **4**A to **4**D are a flowchart showing processing executed by the digital camera **100** in a case where the medical mode is ON. Processing of the present flowchart is executed in a case where the power source of the digital camera **100** has been turned ON in a state where the medical mode is set to be ON, or in a case where the user has switched the setting of the medical mode from OFF to ON in a state where the power source of the digital camera **100** is ON. Processing of each step in the present flowchart is realized by the system control unit **50** deploying the program stored in the nonvolatile memory **56** to the system memory **52** and executing the program, unless specifically stated otherwise. In the following description, the user (photographer) is, for example, a doctor, a nurse, or the like.

[0096] In S**401**, the system control unit **50** turns ON a power saving flag.

[0097] In S**402**, the system control unit **50** connects to an external device via the communication unit **54**. The mode of connection is assumed to be wireless connection, but may be wired connection.

[0098] In S**403**, the system control unit **50** obtains display setting information for the medical information display (FIG. **3**A) from the external device. The display setting information designates display or non-display with respect to each of photographer information (reference signs **302** and **303**), patient information (reference signs **304** and **305**), and area information (reference signs **306** and **307**) on the medical information display.

[0099] In S**404**, the system control unit **50** obtains an area list from the external device. The area list includes candidates for areas that can be selected by the user in the area information obtainment mode.

[0100] In S**405**, the system control unit **50** sets the LV display setting to the medical information display. Therefore, the display format shown in FIG. **3**A is used as an initial display format of the LV screen.

[0101] In S**406**, the system control unit **50** displays the LV screen corresponding to the LV display setting on the display unit **28**. In a case where the LV display setting is the medical information display, the LV screen shown in FIG. **3**A is displayed; in a case where the LV display setting is the shooting parameter display, the LV screen shown in FIG. **3**B is displayed.

[0102] In S**407**, the system control unit **50** judges whether a state where the user has not operated the digital camera **100** (an unoperated state) has continued for a predetermined time period. In a case where the unoperated state has continued for the predetermined time period, processing proceeds to S**436**; otherwise, processing proceeds to S**408**.

[0103] In S**408**, the system control unit **50** judges whether the user has selected the patient information frame **305** on the LV screen. The selection of the patient information frame **305** is made by touching the patient information frame **305**, or by moving the selection frame to the patient information frame **305** through a dial operation or a direction key operation and depressing the SET button **75**. In a case where the patient information frame **305** has been selected, processing proceeds to S**409**; otherwise, processing proceeds to S**410**.

[0104] In S**409**, the system control unit **50** executes processing in the patient information obtainment mode. Through this processing, medical information (here, patient information) to be associated with a lesion image is set. The details of processing of S**409** will be described later with reference to FIGS. **5**A to **5**C and FIGS. **6**A and **6**B.

[0105] In S**410**, the system control unit **50** judges whether the user has selected the photographer information frame **303** on the LV screen. The selection of the photographer information frame **303** is made by touching the photographer information frame **303**, or by moving the selection frame to the photographer information frame **303** through a dial operation or a direction key operation and depressing the SET button **75**. In a case where the photographer information frame **303** has been selected, processing proceeds to S**411**; otherwise, processing proceeds to S**412**.

[0106] In S**411**, the system control unit **50** executes processing in the photographer information obtainment mode. Through this processing, medical information (here, photographer information) to be associated with a lesion image is set. The details of processing of S**411** will be described later with reference to FIGS. **5**A to **5**C and FIGS. **6**A and **6**B.

[0107] In S**412**, the system control unit **50** judges whether the user has selected the area information frame **307** on the LV screen. The selection of the area information frame **307** is made by touching the area information frame **307**, or by moving the selection frame to the area information frame **307** through a dial operation or a direction key operation and depressing the SET button **75**. In a case where the area information frame **307** has been selected, processing proceeds to S**413**; otherwise, processing proceeds to S**414**.

[0108] In S**413**, the system control unit **50** executes processing in the area information obtainment mode. Through this processing, medical information (here, area information) to be associated with a lesion image is set. The details of processing of S**413** will be described later with reference to FIGS. **7**A and **7**B and FIG. **8**.

[0109] With the completion of setting of the medical information through processing of S**409**, S**411**, or S**413**, the system control unit **50** turns OFF the power saving flag in S**430**. Therefore, according to processing in the medical mode of the present embodiment (FIGS. **4**A to **4**D), a power saving function is disabled upon obtainment and setting of at least a part of the medical information.

[0110] In S**414**, the system control unit **50** determines whether the user has depressed the display switching button **83**. In a case where the display switching button **83** has been depressed, processing proceeds to S**415**; otherwise, processing proceeds to S**418**.

[0111] In S**415**, the system control unit **50** determines whether the LV display setting is the medical information display. In a case where the LV display setting is the medical information display, processing proceeds to S**416**; otherwise, processing proceeds to S**417**.

[0112] In S**416**, the system control unit **50** sets (changes) the LV display setting to the shooting parameter display.

[0113] In S**417**, the system control unit **50** sets (changes) the LV display setting to the medical information display.

[0114] In S**418**, the system control unit **50** determines whether the user has performed a dial operation. In a case where the user has performed a dial operation, processing proceeds to S**419**; otherwise, processing proceeds to S**422**.

[0115] In S**419**, the system control unit **50** determines whether the LV display setting is the medical information display. In a case where the LV display setting is the medical information display, processing proceeds to S**420**; otherwise, processing proceeds to S**421**.

[0116] In S**420**, the system control unit **50** moves the selection frame for the medical information on the LV screen corresponding to the medical information display (FIG. **3**A). Note that in a case where the dial operation has been performed during half-pressing of the shutter button **61**, the system control unit **50** may disable the movement of the selection frame for the medical information.

[0117] In S**421**, the system control unit **50** changes a shooting parameter corresponding to the operated dial. The digital camera **100** has, for example, a shutter speed, an f-number, exposure correction, and ISO as shooting parameters. The shooting parameters are assigned on a per-dial basis, and this assignment may vary depending on a shooting mode.

[0118] In S**422**, the system control unit **50** determines whether the user has half-pressed the shutter button **61**. In a case where the user has half-pressed the shutter button **61**, processing proceeds to S**423**; otherwise, processing proceeds to S**427**.

[0119] In S**423**, the system control unit **50** sets the LV display setting to the medical information display.

[0120] In S**424**, the system control unit **50** displays the LV screen corresponding to the LV display setting on the display unit **28**. As the LV display setting has been set to the medical information display in S**423**, the LV screen corresponding to the medical information display (FIG. **3**A) is displayed in S**424**.

[0121] In S**425**, the system control unit **50** determines whether the user has full-pressed the shutter button. In a case where the user has full-pressed the shutter button, processing proceeds to S**426**; otherwise, processing returns to S**422**.

[0122] In S**426**, the system control unit **50** executes processing in the lesion shooting mode. The details of processing of S**426** will be described later with reference to FIG. **9**.

[0123] In S**427**, the system control unit **50** judges whether the user has selected the end button **308** on the LV screen. The selection of the end button **308** is made by touching the end button **308**, or by moving the selection frame to the end button **308** through a dial operation or a direction key operation and depressing the SET button **75**. Alternatively, as another example, the system control unit **50** may determine that the end button **308** has been selected in a case where an end button function is assigned to a button included in other operation members **70***b* and the pertinent button has been pressed. In a case where the end button **308** has been selected, processing proceeds to S**428**; otherwise, processing proceeds to S**431**.

[0124] In S**428**, the system control unit **50** executes processing in the lesion shooting end mode. As a result of this processing, the setting of the medical information that has been set to be associated with the lesion image is cancelled. The setting of the medical information is cancelled by, for example, clearing (deleting) the medical information held in the system memory **52**. The details of processing of S**428** will be described later with reference to FIG. **10** and FIGS. **11**A and **11**B.

[0125] In S**429**, the system control unit **50** turns ON the power saving flag. Therefore, according to processing in the medical mode of the present embodiment (FIGS. **4**A to **4**D), the power saving function is enabled upon cancellation of the setting of the medical information.

[0126] In S**431**, the system control unit **50** judges whether the medical mode is to be ended. For example, it is determined that the medical mode is to be ended in a case where the user has switched the medical mode to OFF, or the user has instructed to turn OFF the power source of the digital camera **100** by operating the power source switch **72**. In a case where the medical mode is to be ended, processing proceeds to S**432**; otherwise, processing returns to S**406**.

[0127] In S**432**, the system control unit **50** cancels the setting of the medical information. As stated earlier, the setting of the medical information is cancelled by, for example, clearing (deleting) the medical information held in the system memory **52**.

[0128] In S**433**, the system control unit **50** turns ON the power saving flag. Therefore, according to processing in the medical mode of the present embodiment (FIGS. **4**A to **4**D), the setting of the medical information is cancelled and the power saving function is enabled upon ending the medical mode (switching the medical mode from ON to OFF).

[0129] In S**436**, the system control unit **50** determines whether the power saving flag is ON. In a case where the power saving flag is ON, processing proceeds to S**437**; otherwise, processing proceeds to S**438**.

[0130] In S**437**, the system control unit **50** executes power saving processing (processing for causing the digital camera **100** to transition to a power saving state). For example, the system control unit **50** executes processing for turning OFF the power source of the digital camera **100** as the power saving processing. Alternatively, the power saving processing may be one of the processes indicated below, or a combination of two or more processes included among the processes indicated below.

- **[0131]** Turning OFF the display unit **28**
- **[0132]** Reducing the luminance of the display unit **28**
- **[0133]** Reducing the LV display rate of the display unit **28**
- **[0134]** Reducing the LV display resolution of the display unit **28**
- **[0135]** Stopping a communication operation of the communication unit **54**
- **[0136]** Turning OFF the image capturing unit **22**
- **[0137]** Turning OFF the EVF **29**

[0138] Note that in a case where the display unit **28** has been turned OFF or reduced in luminance through the power saving processing, there is a possibility that the user mistakenly considers that the power source of the digital camera **100** has been turned OFF, and turns OFF the power source of the digital camera **100** by operating the power source switch **72**. To avoid the occurrence of this situation, the system control unit **50** may disable an operation on the power source switch **72**.

[0139] Furthermore, the digital camera **100** may be configured so that the user can set the contents of the power saving processing. A setting screen for making this setting will be described later with reference to FIGS. **13**A and **13**B.

[0140] In S**438**, the system control unit **50** provides a notification that prompts the user to select whether to end shooting. This notification is provided by, for example, displaying a message that inquires of the user whether to end shooting (an end-of-shooting confirmation message).

[0141] In S**439**, the system control unit **50** determines whether the user has instructed (selected) to end shooting. For example, in S**438**, an "end shooting" button and a "continue shooting" button are displayed together with the end-of-shooting confirmation message; in a case where the user has selected the "end shooting" button, the system control unit **50** determines that the user has instructed (selected) to end shooting. In a case where the user has selected the "continue shooting" button, the system control unit **50** determines that the user has not instructed (selected) to end shooting. In a case where the user has instructed to end shooting, processing proceeds to S**428**; otherwise, processing returns to S**406**.

[0142] Note that the external system that communicates with the digital camera **100** may include an electronic medical record terminal, although the illustration thereof is omitted. In this case, the system control unit **50** may detect that a display target patient for whom an electronic medical record is to be displayed on the electronic medical record terminal has been changed from a current patient (a first patient) to another patient (a second patient) by communicating with the external system. In response to the detection of the change in the display target patient, the system control unit **50** may cancel the setting of the medical information and turn ON the power saving flag.

[0143] Furthermore, the system control unit **50** may detect the end of hours of medical practice in a medical facility that uses the digital camera **100** by communicating with the external system. For example, the digital camera **100** can obtain medical practice schedule information, which includes the ending time of hours of medical practice, from the external device, and detect the end of hours of medical practice by judging whether the current time is past the ending time. In a case where the end of hours of medical practice has been detected, the system control unit **50** may cancel the setting of the medical information and turn ON the power saving flag by executing processing in the lesion shooting end mode.

[0144] Furthermore, in S**429** and S**433**, the system control unit **50** may judge whether a time period to the starting time of the medical practice for the next patient is shorter than a threshold by communicating with the external system. For example, the system control unit **50** can obtain medical practice schedule information from the external device, and judge whether the time period to the starting time of the medical practice for the next patient is shorter than the threshold on the basis of the current time and the medical practice schedule information. In a case were the time period to the starting time of the medical practice for the next patient is shorter than the threshold, the system control unit **50** may keep the power saving flag OFF, without turning ON the same. That is to say, even in a case where the medical information has not been set (a case where the setting of the medical information has been cancelled), if the time period to the starting time of the medical practice for the next patient is shorter than the threshold, the system control unit **50** may perform control so as not to cause the digital camera

100 to transition to the power saving state, even when the unoperated state has continued for the predetermined time period.

Processing in Patient Information Obtainment Mode and Photographer Information Obtainment Mode

[0145] Processing in the patient information obtainment mode and the photographer information obtainment mode will be described with reference to FIGS. **5**A to **5**C and FIGS. **6**A and **6**B. The following description, which will be provided with reference to FIGS. **5**A to **5**C and FIGS. **6**A and **6**B, applies to both of processing in the patient information obtainment mode and processing in the photographer information obtainment mode, unless specifically stated otherwise.

[0146] FIG. **5**A is a diagram showing an example of a barcode obtainment screen displayed in the patient information obtainment mode and the photographer information obtainment mode.

[0147] **501** indicates a display region of an LV image of a subject. In the display region **501**, in order to emphasize a region **511** in which a barcode should be located for shooting, a portion other than the region **511** is displayed in a dark state.

[0148] **502** is a MENU button. In a case where the user wishes to exit the barcode obtainment screen (to end the patient information obtainment mode or the photographer information obtainment mode), the user selects the MENU button **502**.

[0149] **512** is a message that prompts the user to shoot the barcode. Note that the message **512** may include information indicating whether the current mode is the patient information obtainment mode or the photographer information obtainment mode. For example, in the case of the patient information obtainment mode, the message **512** may be "please shoot the barcode of the patient", whereas in the case of the photographer information obtainment mode, the message **512** may be "please shoot the barcode of the photographer".

[0150] FIG. **5**B is a diagram showing an example of a confirmation screen of patient information (in the case of the patient information obtainment mode) or photographer information (in the case of the photographer information obtainment mode) corresponding to the shot barcode. The patient information and the photographer information are received from an external device. **503** is a region that displays the patient information or the photographer information received from the external device. In the examples of FIGS. **5**A to **5**C, a full name of a patient or a full name of a photographer included in the received patient information or photographer information is displayed.

[0151] **504** is an OK button. In a case where information displayed in the region **503** is correct, the user selects the OK button **504**.

[0152] **505** is a re-shoot button **505**. In a case where the user wishes to re-shoot the barcode, the user selects the re-shoot button **505**.

[0153] FIG. **5**C is a diagram showing an example of an error screen displayed in a case where the obtainment (reception) of the patient information or the photographer information has failed.

[0154] **506** is a message announcing an error. The displayed message may be changed in accordance with the content of the error.

[0155] **507** is an end button. In a case where the user is to give up the obtainment of the patient information or the photographer information, the user selects the end button **507**.

[0156] **508** is a re-shoot button **508**. In a case where the user wishes to re-shoot the barcode in an attempt to obtain the patient information or the photographer information again, the user selects the re-shoot button **508**.

[0157] FIGS. **6**A and **6**B are flowcharts of processing in the patient information obtainment mode and the photographer information obtainment mode.

[0158] In S**601**, the system control unit **50** displays the barcode obtainment screen shown in FIG. **5**A on the display unit **28**.

[0159] In S**602**, the system control unit **50** judges whether a cancel instruction has been issued. The system control unit **50** determines that the cancel instruction has been issued in a case where the MENU button **502** has been selected on the barcode obtainment screen, or in a case where the menu button **81** has been depressed. In a case where the cancel instruction has been issued, the patient information obtainment mode or the photographer information obtainment mode is ended, and processing proceeds to S**410** (in the case of the patient information obtainment mode) or S**412** (in the case of the photographer information obtainment mode). In a case where the cancel instruction has not been issued, processing proceeds to S**603**.

[0160] In S**603**, the system control unit **50** determines whether the shutter button **61** has been full-pressed. In a case where the shutter button **61** has been full-pressed, processing proceeds to S**604**; otherwise, processing returns to S**602**.

[0161] In S**604**, the system control unit **50** shoots a barcode, and transmits an obtained barcode image to an external device via the communication unit **54**. Alternatively, the system control unit **50** may obtain a code by analyzing the barcode image, and transmit obtained code information to the external device via the communication unit **54**.

[0162] In S**605**, the system control unit **50** judges whether a response has been received from the external device. The system control unit **50** repeats the judgment of S**605** until the response is received; once the response has been received, processing proceeds to **606**.

[0163] In S**606**, the system control unit **50** determines whether the received response is normal. The system control unit **50** determines that the response is abnormal in a case where medical information (patient information or photographer information) in the response is corrupted or the response does not include intended medical information. In a case where the response is normal, processing proceeds to S**607**; otherwise, processing proceeds to S**608**.

[0164] In S**607**, based on the patient information or the photographer information included in the received response, the system control unit **50** displays the confirmation screen shown in FIG. **5**B on the display unit **28**.

[0165] In S**608**, the system control unit **50** displays an error screen shown in FIG. **5**C on the display unit **28**.

[0166] In S**609**, the system control unit **50** judges whether an end instruction has been issued. The system control unit **50** determines that the end instruction has been issued in a case where the user has selected the end button **507** on the error screen, or in a case where the user has moved a selection frame to the end button **507** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the end instruction has been issued, the patient information obtainment mode or the photographer information obtainment mode is ended, and processing proceeds to S**410** (in the case of the patient information obtainment mode) or S**412** (in the case of the photographer information obtainment mode). In a case where the end instruction has not been issued, processing proceeds to S**610**.

[0167] In S**610**, the system control unit **50** judges whether a re-shoot instruction has been issued. The system control unit **50** determines that the re-shoot instruction has been issued in a case where the user has selected the re-shoot button **508** on the error screen, or in a case where the user has moved the selection frame to the re-shoot button **508** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the re-shoot instruction has been issued, processing returns to S**601**; otherwise, processing returns to S**609**.

[0168] In S**611**, the system control unit **50** judges whether the user has issued an OK instruction. The system control unit **50** determines that the OK instruction has been issued in a case where the user has selected the OK button **504** on the confirmation screen, or in a case where the user has moved the selection frame to the OK button **504** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the OK instruction has been issued, processing proceeds to S**613**; otherwise, processing proceeds to S**612**.

[0169] In S**612**, the system control unit **50** judges whether a re-shoot instruction has been issued. The system control unit **50** determines that the re-shoot instruction has been issued in a case where the user has selected the re-shoot button **505** on the confirmation screen, or in a case where the user has moved the selection frame to the re-shoot button **505** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the re-shoot instruction has been issued, processing returns to S**601**; otherwise, processing returns to S**611**.

[0170] In S**613**, the system control unit **50** holds (stores) the medical information (patient information in the case of the patient information obtainment mode, and photographer information in the case of the photographer information obtainment mode) included in the response received in S**605** in the system memory **52**. As a result, the medical information (patient information or photographer information) to be associated with a lesion image is set. Furthermore, in a case where the LV screen of FIG. **3**A is displayed thereafter, the full name of the photographer is displayed in the photographer information frame **303** (in the case of the photographer information obtainment mode), and the full name of the patient is displayed in the patient information frame **305** (in the case of the patient information obtainment mode).

Processing in Area Information Obtainment Mode

[0171] Processing in the area information obtainment mode will be described with reference to FIGS. **7**A and **7**B and FIG. **8**.

[0172] FIG. **7**A is a diagram showing an example of an area information selection screen. **701** is an area list obtained in S**404**. **702** is a MENU button. In a case where the user wishes to exit the area information selection screen (to end the area information obtainment mode), the user selects the MENU button **702**. FIG. **7**B is a diagram showing an example of an error screen. **703** is a message announcing an error. The displayed message may be changed in accordance with the content of the error. **704** is an end button. When the user has selected the end button **704**, the area information obtainment mode is ended.

[0173] FIG. **8** is a flowchart of processing in the area information obtainment mode. In S**801**, the system control unit **50** determines whether it has been able to obtain an area list from an external device in S**404**. In a case where it has been able to obtain the area list, processing proceeds to S**802**; otherwise, processing proceeds to S**803**.

[0174] In S**802**, the system control unit **50** displays the area information selection screen shown in FIG. **7**A on the display unit **28**.

[0175] In S**803**, the system control unit **50** displays the error screen shown in FIG. **7**B on the display unit **28**. When the end button **704** has been selected on the error screen, processing proceeds to S**414**.

[0176] In S**804**, the system control unit **50** judges whether a cancel instruction has been issued. The system control unit **50** determines that the cancel instruction has been issued in a case where the MENU button **702** has been selected on the area information selection screen, or in a case where the menu button **81** has been depressed. In a case where the cancel instruction has been issued, the area information obtainment mode is ended, and processing proceeds to S**414**. In a case where the cancel instruction has not been issued, processing proceeds to S**805**.

[0177] In S**805**, the system control unit **50** judges whether an area has been selected from the area list **701**. The user can select an area by touching one of the plurality of areas included in the area list. Alternatively, the user can select an area by moving a selection frame to the position of one of the plurality of areas included in the area list through a dial operation or a direction key operation and depressing the SET button **75**. In a case where an area has been selected, processing proceeds to S**806**; otherwise, processing returns to S**804**.

[0178] In S**806**, the system control unit **50** holds (stores) area information indicating the area selected in S**805** in the system memory **52**. As a result, the area information to be associated with a lesion image is set. Also, in a case where the LV screen of FIG. **3**A is displayed thereafter, the area information is displayed in the area information frame **307**.

Processing in Lesion Shooting Mode

[0179] FIG. **9** is a flowchart of processing in the lesion shooting mode. In S**901**, the system control unit **50** shoots a lesion image and saves the same in the memory **32**.

[0180] In S**902**, the system control unit **50** preview-displays the lesion image on the display unit **28**. As one example, the system control unit **50** may display the set medical information (at least one of the patient information, photographer information, and area information) in such a manner that the set medical information is overlaid on the lesion image that has been preview-displayed. Furthermore, execution and inexecution of the preview display may be switchable via a user setting. Furthermore, a display time period of the preview display may be changeable via a user setting.

[0181] In S**903**, the system control unit **50** associates the set medical information (at least one of the patient information, photographer information, and area information) with the lesion image, and records the lesion image in the recording medium **200**. Although a method of association is not limited in particular, for example, the system control unit

50 can write the medical information into a metadata region or the like of an image file including the lesion image. Alternatively, the system control unit **50** may generate a metadata file including the medical information, and associate an image file including the lesion image with the metadata file.

[0182] In S**904**, the system control unit **50** transmits the lesion image and the medical information associated with the lesion image to an external device via the communication unit **54**.

[0183] In S**905**, the system control unit **50** adds information that identifies the lesion image (e.g., the file name of the image file), as well as a "currently transmitting" status indicating that this lesion image is currently being transmitted, to an image transmission list held in the memory **32**.

Processing in Lesion Shooting End Mode

[0184] Processing in the lesion shooting end mode will be described with reference to FIG. **10** and FIGS. **11**A and **11**B.

[0185] FIG. **10** is a diagram showing an example of a confirmation screen for a lesion image. On the confirmation screen, the user can confirm the lesion image shot in S**901** and the medical information (in the example of FIG. **10**, the full name of the patient included in the patient information) associated with this lesion image in S**903**.

[0186] **1001** is a display region of medical information (in the example of FIG. **10**, the full name of the patient included in the patient information) associated with a lesion image with a "currently transmitting" status in the image transmission list. In a case where there are a plurality of lesion images with the "currently transmitting" status, the display region **1001** may display medical information of the lesion image that was shot first or last, or may display pieces of medical information of all lesion images.

[0187] **1002** is a display region that displays the number of lesion images with the "currently transmitting" status in the image transmission list.

[0188] **1003** is a display region that displays a thumbnail image of a lesion image with the "currently transmitting" status in the image transmission list. In a case where there are a plurality of lesion images with the "currently transmitting" status, the display region **1003** may display a thumbnail image of the lesion image that was shot first or last, or may display thumbnail images of all lesion images.

[0189] **1004** is a button for causing the user to select to transfer the lesion image to the medical system and end lesion shooting.

[0190] **1005** is a button for causing the user to select to end lesion shooting without transferring the lesion image to the medical system.

[0191] FIGS. **11**A and **11**B are flowcharts of processing in the lesion shooting end mode. In S**1101**, the system control unit **50** determines whether there is a lesion image with the "currently transmitting" status in the image transmission list. In a case where there is a lesion image with the "currently transmitting" status, processing proceeds to S**1103**; otherwise, processing proceeds to S**1102**.

[0192] In S**1102**, the system control unit **50** clears the medical information (patient information, photographer information, and area information) held in the system memory **52**. As a result, the setting of the medical information is cancelled. Thereafter, the lesion shooting end mode is ended, and processing proceeds to S**429**.

[0193] In S**1103**, the system control unit **50** displays the confirmation screen for the lesion image shown in FIG. **10**.

[0194] In S**1104**, the system control unit **50** judges whether an instruction for ending lesion shooting without transferring the lesion image has been issued. The system control unit **50** determines that the instruction for ending lesion shooting without transferring the lesion image has been issued in a case where the user has selected the "end without transfer" button **1005** on the confirmation screen, or in a case where the user has moved a selection frame to the "end without transfer" button **1005** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the instruction for ending lesion shooting without transferring the lesion image has been issued, processing proceeds to S**1105**; otherwise, processing proceeds to S**1108**.

[0195] In S**1105**, the system control unit **50** transmits a cancel notification to an external device via the communication unit **54**.

[0196] In S**1106**, the system control unit **50** deletes the "currently transmitting" status and the file name of the corresponding lesion image from the image transmission list.

[0197] In S**1107**, the system control unit **50** clears the medical information (patient information, photographer information, and area information) held in the system memory **52**. As a result, the setting of the medical information is cancelled. Thereafter, the lesion shooting end mode is ended, and processing proceeds to $**429**.

[0198] In S**1108**, the system control unit **50** judges whether an instruction for transferring the lesion image and ending lesion shooting has been issued. The system control unit **50** determines that the instruction for transferring the lesion image and ending lesion shooting has been issued in a case where the user has selected the "transfer and end" button **1004** on the confirmation screen, or in a case where the user has moved a selection frame to the "transfer and end" button **1004** through a dial operation or a direction key operation and depressed the SET button **75**. In a case where the instruction for transferring the lesion image and ending lesion shooting has been issued, processing proceeds to S**1109**; otherwise, processing returns to S**1104**.

[0199] In S**1109**, the system control unit **50** transmits an OK notification to the external device via the communication unit **54**.

[0200] In S**1110**, the system control unit **50** determines whether a transmission result of the lesion image transmitted in S**904** has already been received from the external device. In a case where the transmission result has already been received, processing proceeds to S**1111**; otherwise, processing proceeds to S**1114**.

[0201] In S**1111**, the system control unit **50** determines whether the transmission result received from the external device is normal. In a case where the transmission result is normal, processing proceeds to S**1112**; otherwise, processing proceeds to S**1113**.

[0202] In S**1112**, the system control unit **50** changes the status of the lesion image corresponding to the received transmission result to "transmission complete" in the image transmission list.

[0203] In S**1113**, the system control unit **50** changes the status of the lesion image corresponding to the received transmission result to "transmission failed" in the image transmission list.

[0204] In S**1114**, the system control unit **50** determines whether the transmission result has been received from the external device within a certain time period since the start of transmission of the lesion image to the external device in S**904** (i.e., whether a transmission timeout has occurred). In a case where the transmission result has not been received from the external device within the certain time period (in a case where the transmission timeout has occurred), processing proceeds to S**1115**; otherwise, processing proceeds to S**1116**.

[0205] In S**1115**, the system control unit **50** changes the status of the lesion image for which the transmission timeout has occurred to "transmission failed" in the image transmission list.

[0206] In S**1116**, the system control unit **50** determines whether there is a lesion image with the "currently transmitting" status in the image transmission list. In a case where there is a lesion image with the "currently transmitting" status, processing returns to S**1110**; otherwise, processing proceeds to S**1117**.

[0207] In S**1117**, the system control unit **50** deletes list information (the file name and the status of each lesion image) included in the image transmission list.

[0208] In S**1118**, the system control unit **50** clears the held medical information (patient information, photographer information, and area information). As a result, the setting of the medical information is cancelled. Thereafter, the lesion shooting end mode is ended, and processing proceeds to S**429**.

ON/OFF Switching Processing for Medical Mode

[0209] FIG. **12** is a flowchart of processing for switching between ON and OFF of the medical mode. Processing of the present flowchart is started in a case where the user has depressed the menu button **81**, regardless of whether the medical mode is ON or OFF. Note that while the system control unit **50** is executing processing of the patient information obtainment mode, the photographer information obtainment mode, the area information obtainment mode, the lesion shooting mode, or the lesion shooting end mode, processing of the present flowchart is not started even if the menu button **81** has been depressed. Furthermore, the system control unit **50** may perform control so that, in a case where the power saving flag is OFF, processing of the present flowchart is not started even if the menu button **81** has been depressed. Processing of each step in the present flowchart is realized by the system control unit **50** deploying the program stored in the nonvolatile memory **56** to the system memory **52** and executing the program, unless specifically stated otherwise.

[0210] In S**1200**, the system control unit **50** displays a menu screen on the display unit **28**.

[0211] In S**1201**, the system control unit **50** judges whether the user has selected an item of the medical mode on the menu screen. The user can select an item of the medical mode by touching the item of the medical mode on the menu screen. Alternatively, the user can select an item of the medical mode by moving a selection frame to the position of the item of the medical mode through a dial operation or a direction key operation and depressing the SET button **75**. In a case where an item of the medical mode has been selected, the system control unit **50** displays ON and OFF as a sub item of the item of the medical mode, and processing proceeds to S**1202**. In a case where an item of the medical mode has not been selected, processing proceeds to S**1206**.

[0212] In S**1202**, the system control unit **50** judges whether the user has selected ON in the sub item of the item of the medical mode. The user can select ON by touching ON in the sub item. Alternatively, the user can select ON by moving a selection frame to the position of ON through a dial operation or a direction key operation and depressing the SET button **75**. In a case where ON has been selected, processing proceeds to $**1203**; otherwise, processing proceeds to S**1204**.

[0213] In S**1203**, the system control unit **50** sets the medical mode to ON (e.g., changes the operation mode from the medical mode to a normal mode).

[0214] In S**1204**, the system control unit **50** judges whether the user has selected OFF in the sub item of the item of the medical mode. The user can select OFF by touching OFF in the sub item. Alternatively, the user can select OFF by moving a selection frame to the position of OFF through a dial operation or a direction key operation and depressing the SET button **75**. In a case where OFF has been selected, processing proceeds to S**1205**; otherwise, processing returns to S**1202**.

[0215] In S**1205**, the system control unit **50** sets the medical mode to OFF. Once the medical mode has been set to OFF, for example, the operation mode is changed from the medical mode to the normal mode.

[0216] In S**1206**, the system control unit **50** judges whether an instruction for ending a menu has been issued. For example, the system control unit **50** determines that the instruction for ending the menu has been issued in a case where the user has touched a menu end button on the menu screen. In a case where the instruction for ending the menu has been issued, processing of the present flowchart is ended. In a case where the instruction for ending the menu has not been issued, processing returns to S**1201**.

Examples of Setting Screens of Power Saving Processing

[0217] Examples of setting screens of the power saving processing according to the present embodiment will be described with reference to FIGS. **13**A and **13**B. The system control unit **50** can display, as the setting screens, a list display screen of power saving items and a detailed setting screen of power saving items.

[0218] FIG. **13**A is a diagram showing an example of the list display screen of power saving items. **1301** is a display region of the power saving items. "Low-luminance display on monitor" corresponds to processing for "reducing the luminance of the display unit **28**" described in S**437**. Also, in order to improve the power saving performance, "low-luminance display on monitor" may include processing for "reducing the LV display rate of the display unit **28**" and processing for "reducing the LV display resolution of the display unit **28**" described in S**437**. Alternatively, the display region **1301** may include individual items in connection with the processing for "reducing the LV display rate of the display unit **28**" and the processing for "reducing the LV display resolution of the display unit **28**", respectively.

[0219] "Monitor OFF" corresponds to processing for "turning OFF the display unit **28**" described in S**437**. Also, in order to improve the power saving performance, "monitor OFF" may include processing for "turning OFF the image capturing unit **22**" described in S**437**. Alternatively, the display region **1301** may include an individual item corresponding to the processing for "turning OFF the image capturing unit **22**".

[0220] "Communication OFF" corresponds to processing for stopping a communication operation of the communication unit **54**" described in S**437**. For example, the system control unit **50** can stop the communication operation of the communication unit **54** by stopping a power supply to the communication unit **54**. "Viewfinder OFF" corresponds to processing for "turning OFF the EVF **29**" described in S**437**.

[0221] "Auto power OFF" corresponds to processing for turning OFF the power of the entire system of the digital camera **100** (processing for "turning OFF the power source of the digital camera **100**" described in S**437**).

[0222] **1302** is a display region that displays a set time period of each power saving item. As shown in FIG. **13**A, the user can set different time periods for different power saving items. Therefore, the judgment of S**407** is made for each power saving item. Also, in making the judgment for each power saving item, the time periods that have been set for the respective power saving items are used as the "predetermined time period".

[0223] **1303** is a selection frame indicating which item is selected from among the plurality of power saving items that are displayed. The user can move the selection frame **1303** through a dial operation or a direction key operation, and select any power saving item by depressing the SET button **75**.

[0224] FIG. **13**B is a diagram showing an example of a detailed setting screen of the power saving item corresponding to "communication OFF". When "communication OFF" has been selected on the list display screen of FIG. **13**A, the detailed setting screen of FIG. **13**B is displayed.

[0225] **1304** is a display region that displays the power saving item corresponding to the detailed setting screen that is displayed.

[0226] **1305** is a display region that displays a list of settable values (time periods).

[0227] **1306** is a selection frame indicating which value is selected from among the plurality of settable values (time periods) that are displayed. The user can move the selection frame **1306** through a dial operation or a direction key operation, and select any value by depressing the SET button **75**. When the user has selected a specific value, the selected value is set as a time period of "communication OFF".

[0228] Note that the system control unit **50** may perform control to prohibit changing of settings of the power saving processing in a case the power saving flag is OFF (or in a case where medical information is held in the system memory **52**). In this case, changing of the "predetermined time period" used in S**407** is prohibited.

Summary of First Embodiment

[0229] As described above, according to the first embodiment, the digital camera **100** sets specific information (e.g., medical information). The digital camera **100** associates the set specific information with an image shot using the image capturing unit **22** (e.g., a lesion image, which is one example of a medical image). In a case where the specific information has not been set, the digital camera **100** performs control so that the digital camera **100** transitions to the power saving state when the unoperated state has continued for a predetermined time period. Also, in a case where the specific information has been set, the digital camera **100** performs control so that the digital camera **100** does not transition to the power saving state even if the unoperated state has continued for the predetermined time period.

[0230] In a state where specific information to be associated with a shot image has been set, it is considered that the possibility of the user performing shooting in a short while is relatively high. In this state, the digital camera **100** performs control so that the digital camera **100** does not transition to the power saving state even if the unoperated state has continued for the predetermined time period. Therefore, the present embodiment reduces the possibility that the digital camera **100** transitions to the power saving state against the user's intention.

[0231] Conversely, in a state where specific information to be associated with a shot image has not been set, it is considered that the possibility of the user performing shooting in a short while is relatively low. In this state, the digital camera **100** performs control so that the digital camera **100** transitions to the power saving state when the unoperated state has continued for the predetermined time period. Therefore, the present embodiment suppresses an increase in a wasteful power consumption.

[0232] As described above, the present embodiment can reduce the possibility that an image capturing apparatus transitions to the power saving state against the user's intention while suppressing an increase in a wasteful power consumption.

[0233] Note that each type of processing that has been described above as being executed by the system control unit **50** may be executed by one item of hardware, or may be executed by a plurality of items of hardware (e.g., a plurality of processors and circuits) sharing the processing.

[0234] Also, although the present embodiment has been described using an exemplary case where the digital camera **100** is used as an image capturing apparatus, any device can be used as an image capturing apparatus of the present embodiment as long as it is a device including image capturing means. That is to say, the present embodiment is applicable to a mobile terminal like a smartphone, a personal computer, a PDA, a mobile telephone terminal, a mobile image viewer, a display-equipped printer apparatus, a digital photo frame, a music player, a game device, an electronic book reader, and so forth.

[0235] Furthermore, the present embodiment is applicable not only to a body of an image capturing apparatus, but also to a control apparatus that communicates with the image capturing apparatus (including a network camera) via wired or wireless communication and remotely controls the image capturing apparatus. Examples of an apparatus that remotely controls an image capturing apparatus are such apparatuses as a smartphone, a tablet PC, and a desktop PC. An image capturing apparatus can be remotely controlled by providing, from the control apparatus side, a notification of a command for causing the image capturing apparatus to perform various types of operations and settings on the basis of an operation that has been performed on the control apparatus side and processing that has been executed on the control apparatus side. Furthermore, the control apparatus side may be capable of receiving a live-view image shot by the image capturing apparatus via wired or wireless communication and displaying the live-view image.

Other Embodiments

[0236] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0237] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

1. An image capturing apparatus including an image capturing unit, comprising:

a switching unit configured to switch between ON and OFF of a medical mode of the image capturing apparatus, wherein the medical mode is an operation mode for shooting a medical image;

a setting unit configured to set specific information;

an association unit configured to associate the set specific information with an image that has been shot using the image capturing unit; and

a control unit configured to perform, in a case where the medical mode is ON, control to

in a case where the specific information has not been set, cause the image capturing apparatus to transition to a power saving state when a state where a user has not operated the image capturing apparatus has continued for a predetermined time period, and

in a case where the specific information has been set, not cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

2. The image capturing apparatus according to claim **1**, wherein

the image is a medical image of a first patient, and

the specific information includes at least one of information related to the first patient, information related to a photographer, and information related to a shooting target area of the first patient.

3. The image capturing apparatus according to claim **2**, further comprising:

a communication unit; and

a reception unit configured to receive the specific information from an external system via the communication unit.

4. The image capturing apparatus according to claim **3**, further comprising:

a transmission unit configured to transmit, to the external system, the image with which the set specific information has been associated.

5. The image capturing apparatus according to claim **3**, further comprising:

a first detection unit configured to detect, by communicating with the external system via the communication unit, that a display target patient for whom an electronic medical record is to be displayed on an electronic medical record terminal included in the external system has been changed from the first patient to a second patient,

wherein the setting unit cancels the setting of the specific information in response to detection of the change in the display target patient from the first patient to the second patient.

6. The image capturing apparatus according to claim **3**, further comprising:

a second detection unit configured to detect, by communicating with the external system via the communication unit, an end of hours of medical practice in a medical facility that uses the image capturing apparatus,

wherein the setting unit cancels the setting of the specific information in response to detection of the end of hours of medical practice.

7. The image capturing apparatus according to claim **3**, further comprising:

a judgment unit configured to judge, by communicating with the external system via the communication unit, whether a time period to a starting time of medical practice for a next patient is shorter than a threshold,

wherein even in a case where the specific information has not been set, if the time period to the starting time of the medical practice for the next patient is shorter than the threshold, the control unit performs control so as not to cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

8. The image capturing apparatus according to claim **3**, wherein

causing the image capturing apparatus to transition to the power saving state includes stopping a communication operation of the communication unit.

9. The image capturing apparatus according to claim **1**, wherein

the setting unit cancels the setting of the specific information when the user has issued an instruction for ending shooting.

10. The image capturing apparatus according to claim **1**, wherein:

the setting unit sets the specific information in a case where the predetermined operation mode is ON, and cancels the setting of the specific information when the predetermined operation mode has been switched from ON to OFF.

11. The image capturing apparatus according to claim **1**, wherein

the setting unit cancels the setting of the specific information when the user has issued an instruction for turning OFF a power source of the image capturing apparatus.

12. The image capturing apparatus according to claim **1**, wherein

in a case where the specific information has been set, the control unit provides a notification that prompts the user to select whether to end shooting when the state where the user has not operated the image capturing apparatus has continued for the predetermined time period, and

the setting unit cancels the setting of the specific information in a case where the user has selected to end shooting in response to the notification.

13. The image capturing apparatus according to claim **1**, further comprising:

a changing unit configured to change the predetermined time period,

wherein in a case where the specific information has been set, the control unit performs control to prohibit changing of the predetermined time period.

14. A control method executed by an image capturing apparatus including an image capturing unit, comprising:

switching between ON and OFF of a medical mode of the image capturing apparatus, wherein the medical mode is an operation mode for shooting a medical image;

setting specific information;

associating the set specific information with an image that has been shot using the image capturing unit; and

performing, in a case where the medical mode is ON, control to

in a case where the specific information has not been set, cause the image capturing apparatus to transition to a power saving state when a state where a user has not operated the image capturing apparatus has continued for a predetermined time period, and

in a case where the specific information has been set, not cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

15. A non-transitory computer-readable storage medium which stores a program for causing an image capturing apparatus including an image capturing unit to execute a control method comprising:

switching between ON and OFF of a medical mode of the image capturing apparatus, wherein the medical mode is an operation mode for shooting a medical image;

setting specific information;

associating the set specific information with an image that has been shot using the image capturing unit; and

performing, in a case where the medical mode is ON, control to

capturing apparatus to transition to a power saving state when a state where a user has not operated the image capturing apparatus has continued for a predetermined time period, and

in a case where the specific information has been set, not cause the image capturing apparatus to transition to the power saving state even if the state where the user has not operated the image capturing apparatus has continued for the predetermined time period.

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